PAMS: Agile Practices Report

Paragon Apartment Management System

Group 28

Module: UFCF8S-30-2 Advanced Software Development

Assessment: Portfolio (Element 2, Methodology and Contribution Evidence)

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Group Members

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1. Introduction

PAMS (Paragon Apartment Management System) is the desktop application developed for this assessment. The brief describes a UK-based apartment company operating offices in Bristol, Cardiff, London, and Manchester, which is struggling with paper records, weak role separation, and no means of viewing occupancy or finance data across cities. The objective was to replace that infrastructure with a single application usable by five distinct staff roles without compromising data integrity between them.

At the conclusion of the project, the system fulfils every core requirement in the brief. It handles account and user management, tenant registration and lifecycle, apartment records, invoicing and payment, reporting, and maintenance tickets. It enforces role-based access at both the login layer and the sidebar, stores passwords using bcrypt, uses parameterised SQL throughout, and ships with 330 automated tests.

This report does not describe the code. Its purpose is to explain how the four group members worked together to build PAMS, why Scrum was selected, what was delivered in each sprint, how Git was used, and what each member contributed. Evidence in the form of screenshots, git logs, and test output is included in the appendices and referenced from the relevant sections.

2. Methodology Chosen: Scrum

2.1 Why Scrum

Three methodologies were evaluated in the first lab week: Waterfall, Kanban, and Scrum. Waterfall was quickly dismissed: the brief is extensive and it was clear from the first read that not every requirement would be fully understood until some of the system had been built. Locking a design upfront would have necessitated rewriting the plan midway through, which is precisely the scenario Waterfall handles poorly.

Kanban was a better fit in principle, but it assumes a steady flow of tickets rather than a fixed deadline. With a firm submission date (23 April 2026, 14:00 UK), four separable components, and deliverables (UML, code, tests, report, demo) that were more tractable when the calendar was divided into fixed intervals, Scrum provided the appropriate structure.

The module teaching material on Blackboard also covered Scrum in detail, enabling us to consult lecture slides and the lab tutor's feedback when issues arose, rather than inventing process from scratch.

2.2 Sprint Length and Timeline

The project was divided into four sprints of approximately two calendar weeks each. The timeline is summarised in the table below.

Sprint	Dates	Theme
Sprint 1	End Feb to early Mar 2026	Core foundation and role based access
Sprint 2	Mid Mar 2026	Reports, notifications, QA polish

Sprint 3	Late Mar to early Apr 2026	Close brief gaps (edit tenant, notice period, complaints, receipts)
Sprint 4	13 Apr to 23 Apr 2026	UI redesign, PDF export, submission prep

Sprints 1 to 3 were two weeks each. Sprint 4 was deliberately shorter (approximately ten days) so that the final days before the submission deadline were reserved for documentation and rehearsal rather than new code.

2.3 Scrum Roles

Scrum responsibilities were allocated as follows:

- **Scrum Master and Lead Architect: Omar.** Omar managed the Scrum process (sprint planning, project board, pull request reviews, merging to dev) and owned the shared backbone: database layer, authentication, application shell, seed script, test base class, design tokens, and component library. Cross-cutting fixes, resizing, scroll, styling, and the database migration, were routed through him.
- **Product Owner: Feras.** Feras owned the tenant module, the most requirement-intensive part of the system (NI numbers, lease dates, complaints, early termination). The gaps that drove the Sprint 3 plan originated partly from a closer re-read of the tenant sections of the brief.
- **Development Team: all four members.** Every member wrote both code and tests.

2.4 Ceremonies

Four ceremonies were conducted per sprint:

- **Sprint planning.** A single in-person meeting at the start of each sprint, typically at the end of the timetabled lab session. Each one left with a clear task and the agreed interface for their module (method names, return shapes, role permissions) defined in advance to ensure straightforward integration.
- **Daily standup.** Conducted asynchronously via WhatsApp. Each member posted three brief updates every working day: yesterday, today, blockers. This approach was more practical than daily meetings given timetable conflicts on non-lab days.
- **Sprint review.** Held in person on the Friday of the final sprint week. Each feature owner demonstrated their work to the group and, where possible, to the lab tutor. Incomplete items were moved to the next sprint's backlog rather than delaying the review.
- **Sprint retrospective.** Conducted immediately after the review in the same session. The team listed kept, drop, and try items. The retrospective at the end of Sprint 2 produced the Sprint 3 plan, as it was during that meeting that four missing brief requirements were identified (one month notice, complaints log, billing receipts, editing existing tenants).

2.5 Definition of Done

A Definition of Done was agreed in Sprint 1 and maintained throughout the project:

1. Code runs locally with no errors after python3 seed_db.py && python3 main.py.
2. Every new or changed public method has at least one unit test.
3. Tests pass on the reviewer's machine, not only the author's.
4. File header includes the student name and ID of the author.
5. Pull request has been reviewed by at least Omar before merge into dev.
6. No plaintext passwords, no f-string SQL, no messagebox import left over in a redesigned view.

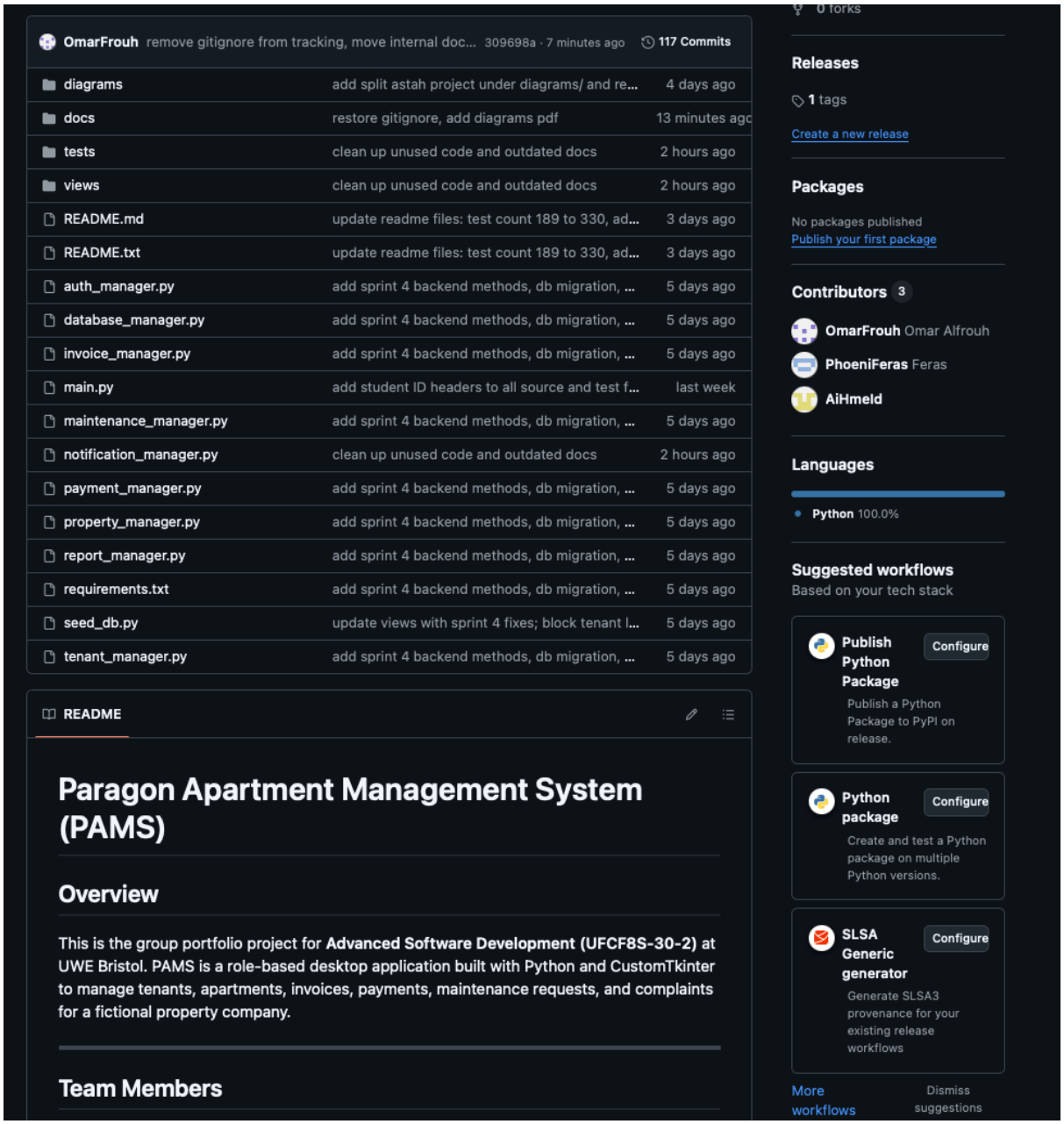
Items 5 and 6 were retrospective additions following issues identified in Sprint 2.

---

3. Tools Used

The toolchain was kept deliberately small. The brief requires a desktop application in an object-oriented language, which we interpreted as a reason to minimise dependencies rather than introduce unnecessary third-party infrastructure.

- **Python 3.13** as the implementation language, consistent with module teaching and providing tkinter as a standard library component on every platform.
- **customtkinter 5.2.2** as the GUI layer over tkinter. This provides the dark theme and themed widgets without requiring a browser or web stack, and maintains the offline constraint implied by the brief's desktop application requirement.
- **SQLite 3** for the database. Built into Python's standard library, zero-install, and file-based so that the marker can inspect pams.db directly. The schema grew to eight tables by Sprint 3.
- **bcrypt 5.0.0** for password hashing. Salted by default, industry standard, and simple to use.
- **fpdf2 2.8.7** for PDF output of invoices and receipts. Added in Sprint 4 when the invoices are wired to save a PDF copy.
- **unittest** (standard library) for automated testing. All 330 tests extend a shared tests/test_base.py:BaseTest class that creates a fresh test database per test.
- **GitHub** for version control and code review via pull requests.
- **Astah Professional** for UML diagrams (use case, class, and four sequence diagrams).
- **Visual Studio Code** as the primary IDE.



4. Sprint Summaries

The four sprints below follow a common template: goal, delivered items, and review note. Detailed per-member contributions for each sprint appear in Section 7.

4.1 Sprint 1: Foundation (End February to Early March 2026)

Goal. Establish the skeleton of every component in the brief at a basic but functional level, with role-based access already enforced.

Delivered:

- Database schema with 7 tables (users, staff, apartments, tenants, maintenance_requests, invoices, payments) and a constant role whitelist.

- Role-aware login and sidebar. Five roles (admin, front_desk, finance_manager, maintenance, manager) each see a different set of tabs.
- Tenant registration with NI number, lease dates, deposit, and monthly rent. Apartment assignment. Early termination with 5% penalty and automatic apartment vacating.
- Invoice generation from tenant monthly rent. Payment recording with atomic transactions (rollback on error). Overdue detection.
- Maintenance ticket lifecycle (Open, Assigned, In Progress, Fixed, Closed). Staff availability check.
- Security fundamentals: bcrypt hashing, parameterised SQL throughout, input validation on every manager method.
- 93 unit tests, all passing.

Review note. The base application worked end to end and the lab tutor Mr. Desmond confirmed in the Week 4 lab that the group was on track. A handful of UI rough edges were logged into the Sprint 2 backlog (small treeviews, no outer scroll on long forms, notification badge showing a count but no detail) rather than patching them in this sprint.

4.2 Sprint 2: Reporting, Notifications and QA (Mid March 2026)

Goal. Add the reporting component called for in the brief, build the notification badge, and resolve the UI defects from the Sprint 1 retrospective.

Delivered:

- report_manager.py with three reports: occupancy by city, financial summary (collected versus pending), and maintenance costs.
- notification_manager.py with overdue invoices, expiring leases (14-day window), upcoming maintenance, and a combined get_notifications() feed.
- Clickable notification bell in the header that opens a popup listing actionable items.
- Tenant module received search, status filter (Active, Archived, Terminated, All), and a separate archived tab.
- Finance module received overdue badges on the invoice list, late payment notifications, monthly revenue by payment method, and a six-month revenue chart.
- Maintenance module received cost reports by apartment and by city, and a notify tenant action recording a scheduled maintenance date.
- UI polish across the application: dark-themed treeviews, automatic column fitting, trackpad scroll support, resizable windows with a sensible minimum size, and outer scrollable frames on every long form.
- 40 additional tests (133 total) covering new managers and edge cases identified during QA.

Review note. At the end of this sprint the team conducted a slow read through of the brief together, which identified that editing tenant records, the one-month notice period, complaint logs, and billing receipts were all explicit requirements not yet delivered. A gap-closure sprint was scoped in response.

4.3 Sprint 3: Brief Gap Closure (Late March to Early April 2026)

Goal. Close the four brief gaps identified in the Sprint 2 retrospective and improve test coverage with edge case scenarios.

Delivered:

- Editing existing tenant records (the brief specifies adding, updating, and removing). Feras added an Edit Tenant tab with lookup by ID and an editable form that prefills from the database.
- One-month notice period on early termination. The existing 5% penalty calculation was retained, but termination now returns a dictionary containing the penalty and the effective date (today plus 30 days) so the view can display both.
- Complaints log with its own table (the eighth in the schema), a Log Complaint tab on the tenant view, and a Resolve action for admin users.
- Billing receipts. Ahmed added get_receipt_data() on PaymentManager and a receipt popup that opens after every successful payment.
- 56 additional tests (189 total), including edge case tests that Numair wrote for complaint isolation, notice period boundaries, and partial tenant updates.

Review note. The sprint was completed on time and every gap was closed before the 12 April cut-off. Three separate pull requests (#26 Feras, #27 Ahmed, #28 Numair) made the merge into dev straightforward because each touched a different module.

4.4 Sprint 4: UI Redesign and Submission Prep (13 to 23 April 2026)

Goal. Produce a submission-quality application: consistent visual design, all submission deliverables complete, and no last-day rush.

Delivered (UI redesign, merged 17 April):

- views/theme.py holds every colour, font, spacing, and radius constant. Updating a token propagates throughout the entire application.
- Sidebar with active-state highlight (3-pixel primary left border plus muted fill), header with a brand block and dynamic page title, and a notification dropdown replacing the Sprint 2 popup.
- Toast notifications replace every messagebox.showinfo/showerror/showwarning call, and a custom confirm_dialog replaces askyesno. All dialogs now honour the dark theme.
- Dashboard on first login with KPI tiles (apartments, tenants, tickets, revenue), recent activity, and alerts.
- Status and priority columns in every table render as coloured pills (Active, Overdue, Urgent, etc.) via tag swapping.

Delivered (backend and tests):

- views/pdf_generator.py using fpdf2 produces branded invoice and receipt PDFs.
- Database migration (_apply_migrations) runs on every boot and idempotently removes a legacy UNIQUE constraint on tenants.apartment_id so a tenant can be reactivated into a previously occupied unit.

- Tenant login blocked at the login screen with a clear staff-only system message. The tenant role still exists as FK infrastructure but cannot be used to sign in.
- Tenant reactivation, invoice search by status and tenant name, and CSV export on the report module.
- 9 new test files raise coverage from 189 to 330 tests, all passing.
- Merge dev into main; tag submission-final.
- Regenerate pams.db from seed_db.py immediately before zipping.
- Build the submission ZIP.
- Final demo rehearsal.

Review note. Sprint 4 was the most ambitious on paper because a full UI redesign was appended to what was intended as submission prep. It succeeded because nothing in *_manager.py, database_manager.py, auth_manager.py, or tests/* was changed, so the 189 existing tests continued passing while the view layer was rewritten. The redesign was isolated on feature/sprint4-redesign and merged to dev only once test count was verified.

5. Git Workflow in Practice

The branching model was simple and unchanged after Sprint 1:

```
main                  (protected, only Omar merges, final stable line)
 └── dev              (integration branch, every feature lands here first)
      └── feature/*   (per person, per feature)
```

Enforced rules:

- Nobody pushed directly to main. Only Omar merged dev into main at the end of a sprint.
- Every feature lived on its own branch named descriptively, e.g. feature/sprint3-missing-features or feature/sprint4-redesign.
- Every branch had a pull request and at least one review before merge, even for single-person features.

5.1 Sprint 3 Pull Requests

PR	Author	Content
#26	Feras	Edit Tenant tab, 1 month notice period, complaint log methods
#27	Ahmed	Billing receipt popup, get_receipt_data, record_payment returns payment_id
#28	Numair	16 edge case tests across complaint isolation, update field targeting, notice period boundaries

5.2 Sprint 4 Merges

- PR #29 (Omar): updated README.md, README.txt, and requirements.txt.
- feature/sprint4-redesign (Omar plus team backend additions): UI redesign plus backend and test expansion. Merged into dev on 17 April as commit e092958.
- diagrams/ folder added on 18 April (commit ca7f9a4) with the three split Astah project files and their PNG and SVG exports.

```
Showing 7 of 7 pull requests in OmarFrouh/ASD-UWE-2026 that match your search

ID   TITLE                                                          BRANCH                      CREATED AT
#28  Sprint 3 (Numair): edge-case tests for complaints, update tenant, not…  feature/testing-sprint3     about 10 days ago
#27  Sprint 3 (Ahmed): billing receipt popup after payment, get_receipt_da…  feature/finance-sprint3     about 13 days ago
#26  Implement Sprint 3 features: edit tenant, early termination dictionar…  feature/tenant-sprint3      about 14 days ago
#25  Sprint 2 (Ahmed): financial summary, late payment notifications, over…  feature/finance-sprint2     about 16 days ago
#24  Sprint 2 (Feras): tenant search, filter by status, summary bar          feature/tenant-sprint2      about 18 days ago
#23  Sprint 1 (Ahmed 24046227): invoice generation, payment recording, ove…  feature/finance-module      about 23 days ago
#22  Sprint 1 (Feras): Tenant registration, list, early termination, archive  feature/tenant-management  about 23 days ago
```

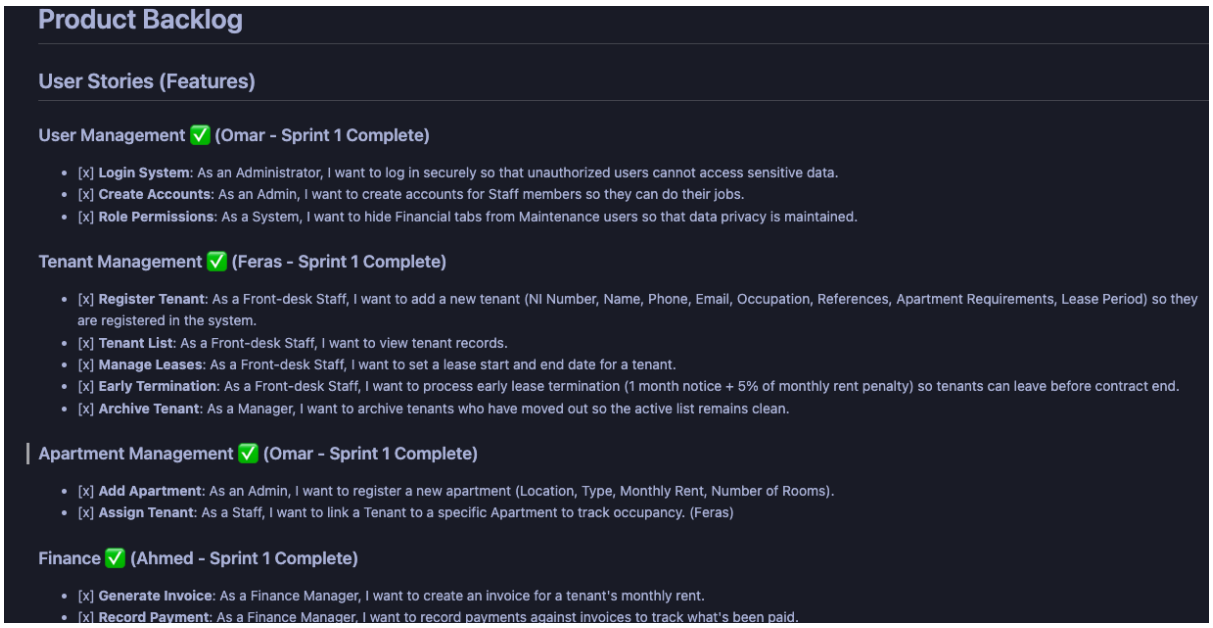
```
(venv) → ASD-UWE-2026 git:(main) ✗ git log --oneline --graph --all | head -40
* 309698a remove gitignore from tracking, move internal docs to archive
* 169d173 restore gitignore, add diagrams pdf
*   d98f054 merge dev into main: testing document and final fixes
|\
| * b3585a6 add testing document (element 3)
| * 89f2a67 clean up unused code and outdated docs
| * 9b529d8 fix invoice pdf export crash on non-latin-1 characters in tenant fields
* | 2ad8d56 merge cleanup and doc refresh into main
|\|
| * 711c9d5 clean up repo and refresh planning docs
* | 606ffeb merge dev into main: sprint 4 final submission
|\|
| * dc1573b update readme files: test count 189 to 330, add fpdf2 dependency
| * 21b855e move planning docs to docs/ folder
| * 8107f1e remove local dev notes from repo
| * 251ccc2 add split astah project under diagrams/ and remove stale pmas.asta
| *   ebe6a2d merge sprint 4 redesign into dev
| |\
| | * d86d206 update views with sprint 4 fixes; block tenant login; rename test data
| | * 6d925a4 add sprint 4 backend methods, db migration, extended tests, and pdf generator
| | * ab03111 wrap all tabviews in bordered card shell for visual consistency
| | * 6f9e055 fix pack/grid conflict in register tab: use grid-compatible section headers
| | * 8f29dd6 fix form resizability: sticky=ew, weight=1 columns, no fixed entry widths
| | * 8603e8c notification dropdown, keyboard shortcuts, empty states, hover highlight
| | * f6dbcbf redesign tenants view
| | * 9187c8b redesign maintenance view
| | * cdb788b redesign reports view
| | * 24bf321 redesign payments view and receipt popup
| | * f5cb721 redesign invoices view
| | * a4fcd1d redesign users and apartments views
| | * d540bb0 add design tokens and redesign login + main shell with sidebar nav and dashboard
| |/
| *   e7ead64 sprint 4: student ID headers, README, requirements, README.txt
| |\
| | * d17d55e update sprint 4 status - readme.txt done, checklist current
| | * 42d7b8c add README.txt with run instructions and test accounts
| | * 8aa0ba7 add student ID headers to all source and test files
| |/
| *   7035fa3 Merge pull request #28 from OmarFrouh/feature/testing-sprint3
| |\
(venv) → ASD-UWE-2026 git:(main) ✗
```

6. Evidence of Collaboration

Evidence of how the team worked together, rather than only what was produced is presented below and in the appendices.

6.1 Project Backlog: docs/Backlog.md

The product backlog was maintained as a Markdown file (docs/BACKLOG.md) in the team repository, version-controlled alongside the source code. User stories were organised by module and sprint, each prefixed with a checkbox ([ ] / [x]) to indicate completion status.



Product Backlog

User Stories (Features)

User Management ✅ (Omar - Sprint 1 Complete)

- [x] **Login System**: As an Administrator, I want to log in securely so that unauthorized users cannot access sensitive data.
- [x] **Create Accounts**: As an Admin, I want to create accounts for Staff members so they can do their jobs.
- [x] **Role Permissions**: As a System, I want to hide Financial tabs from Maintenance users so that data privacy is maintained.

Tenant Management ✅ (Feras - Sprint 1 Complete)

- [x] **Register Tenant**: As a Front-desk Staff, I want to add a new tenant (NI Number, Name, Phone, Email, Occupation, References, Apartment Requirements, Lease Period) so they are registered in the system.
- [x] **Tenant List**: As a Front-desk Staff, I want to view tenant records.
- [x] **Manage Leases**: As a Front-desk Staff, I want to set a lease start and end date for a tenant.
- [x] **Early Termination**: As a Front-desk Staff, I want to process early lease termination (1 month notice + 5% of monthly rent penalty) so tenants can leave before contract end.
- [x] **Archive Tenant**: As a Manager, I want to archive tenants who have moved out so the active list remains clean.

Apartment Management ✅ (Omar - Sprint 1 Complete)

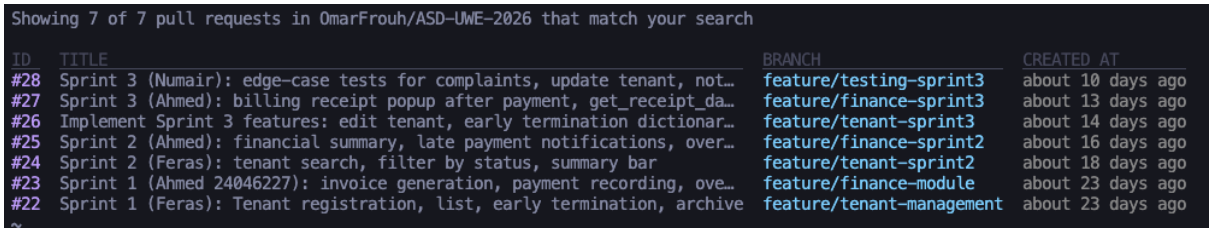
- [x] **Add Apartment**: As an Admin, I want to register a new apartment (Location, Type, Monthly Rent, Number of Rooms).
- [x] **Assign Tenant**: As a Staff, I want to link a Tenant to a specific Apartment to track occupancy. (Feras)

Finance ✅ (Ahmed - Sprint 1 Complete)

- [x] **Generate Invoice**: As a Finance Manager, I want to create an invoice for a tenant's monthly rent.
- [x] **Record Payment**: As a Finance Manager, I want to record payments against invoices to track what's been paid.

6.2 Pull Requests

Every merge into dev went through a pull request. The merged PR list shows reviewer activity on each one.



Showing 7 of 7 pull requests in OmarFrouh/ASD-UWE-2026 that match your search

ID	TITLE	BRANCH	CREATED AT
#28	Sprint 3 (Numair): edge-case tests for complaints, update tenant, not…	feature/testing-sprint3	about 10 days ago
#27	Sprint 3 (Ahmed): billing receipt popup after payment, get_receipt_da…	feature/finance-sprint3	about 13 days ago
#26	Implement Sprint 3 features: edit tenant, early termination dictionar…	feature/tenant-sprint3	about 14 days ago
#25	Sprint 2 (Ahmed): financial summary, late payment notifications, over…	feature/finance-sprint2	about 16 days ago
#24	Sprint 2 (Feras): tenant search, filter by status, summary bar	feature/tenant-sprint2	about 18 days ago
#23	Sprint 1 (Ahmed 24046227): invoice generation, payment recording, ove…	feature/finance-module	about 23 days ago
#22	Sprint 1 (Feras): Tenant registration, list, early termination, archive	feature/tenant-management	about 23 days ago

6.3 Contributor Stats

The command git shortlog -sne shows the number of commits per author across the whole project.

```
   103  Omar Al Frouh <omarfrouh@gmail.com>
     3  Numair <numair@student.uwe.ac.uk>
     3  Omar Alfrouh <omarfrouh@gmail.com>
     2  Ahmed Deraz <asser@Mac.rocket-fibre.home>
     2  Feras <122075868+PhoeniFeras@users.noreply.github.com>
     2  Feras <Feras@ferass-mbp.home>
     1  Ahmed Deraz <asser@MacBook-Air.local>
     1  AiHmeId <ahmed2.deraz@live.uwe.ac.uk>
     1  Feras <Feras@Ferass-MacBook-Pro.local>
~
```

6.4 Automated Test Output

The final test run before writing this report reports 330 tests with zero failures.

```
test_update_terminated_tenant_fails (test_sprint3.TestUpdateTenant.test_update_terminated_tenant_fails) ... ok
test_archive_tenant_clears_apartment_ghost (test_tenant_manager.TestTenantManager.test_archive_tenant_clears_apartment_ghost) ... ok
test_assign_occupied_apartment_fails (test_tenant_manager.TestTenantManager.test_assign_occupied_apartment_fails) ... ok
test_early_termination_charges_penalty (test_tenant_manager.TestTenantManager.test_early_termination_charges_penalty) ... ok
test_register_and_assign_tenant (test_tenant_manager.TestTenantManager.test_register_and_assign_tenant) ... ok
test_reactivate_already_active_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_already_active_raises) ... ok
test_reactivate_archived_becomes_active (test_tenant_reactivate.TestTenantReactivate.test_reactivate_archived_becomes_active) ... ok
test_reactivate_nonexistent_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_nonexistent_raises) ... ok
test_reactivate_terminated_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_terminated_raises) ... ok

----------------------------------------------------------------------
Ran 330 tests in 75.027s

OK
(venv) → ASD-UWE-2026 git:(main) ✗
```

7. Individual Contributions

Each section below is authored by the member it names. File and line references are to the main branch.

7.1 Omar Al Frouh (24020879), Group Leader and Lead Architect

I led the group across all four sprints as Scrum Master and Lead Architect. That meant running the Scrum ceremonies (planning, standups, reviews, retrospectives), maintaining the project backlog, owning the shared backend infrastructure that every other module depends on, and coordinating the submission deliverables at the end.

Sprint 1, Foundation

The project was initialised. This included writing the initial database schema and database_manager.py (8 tables by completion, with PRAGMA foreign_keys, a centralised connection, and an idempotent create_tables method). auth_manager.py was written around bcrypt with a fixed role whitelist, the login and logout flows, and the dashboard statistics method. property_manager.py covered apartment CRUD; seed_db.py provided a one-command reset to a known good state; main.py constructs every manager and passes it to the view layer. On the view side: views/login_view.py, views/main_view.py (role-based sidebar, header, tab switching), views/user_view.py, views/apartment_view.py, and the first version of views/ui_helpers.py (shared helpers for dark Treeviews, scroll, and column autofit). The test infrastructure was established with tests/test_base.py:BaseTest (fresh database per test, clean teardown), along with the first approximately 30 tests exercising the

base infrastructure. Module contracts (method names, return shapes, role permissions) were agreed upfront with each teammate so that integration onto dev was straightforward.

Sprint 2, Reporting, Notifications, QA Lead

I wrote report_manager.py (occupancy by city, financial summary, maintenance cost) and notification_manager.py (overdue invoices, expiring leases, combined get_notifications feed), then built the Reports tab in views/report_view.py with city filter dropdowns and KPI summary cards. I added the clickable notification bell to the header. At the end of the sprint I ran a QA pass that caught several issues: the missing user list on the Users tab (USR-01), auth_manager.get_all_users() and property_manager.get_cities() being absent, a hardcoded city dropdown that should have queried the database, broken resize and fullscreen support, missing CTkScrollableFrame on long form views, and a enable_trackpad_scroll bug that was overriding the Treeview's own scroll handler.

Sprint 3, Schema, Gap Closure, Reviewer

I extended the schema with the complaints table, updated seed_db.py to seed realistic complaint data, and wrote tests/test_sprint3.py with 32 new tests covering the four brief requirements that were still outstanding. I also reviewed and merged all three Sprint 3 PRs: PR #26 (Feras), PR #27 (Ahmed), PR #28 (Numair). For each one I pulled the branch locally, ran the full test suite, read the diff line by line, and returned comments before merging.

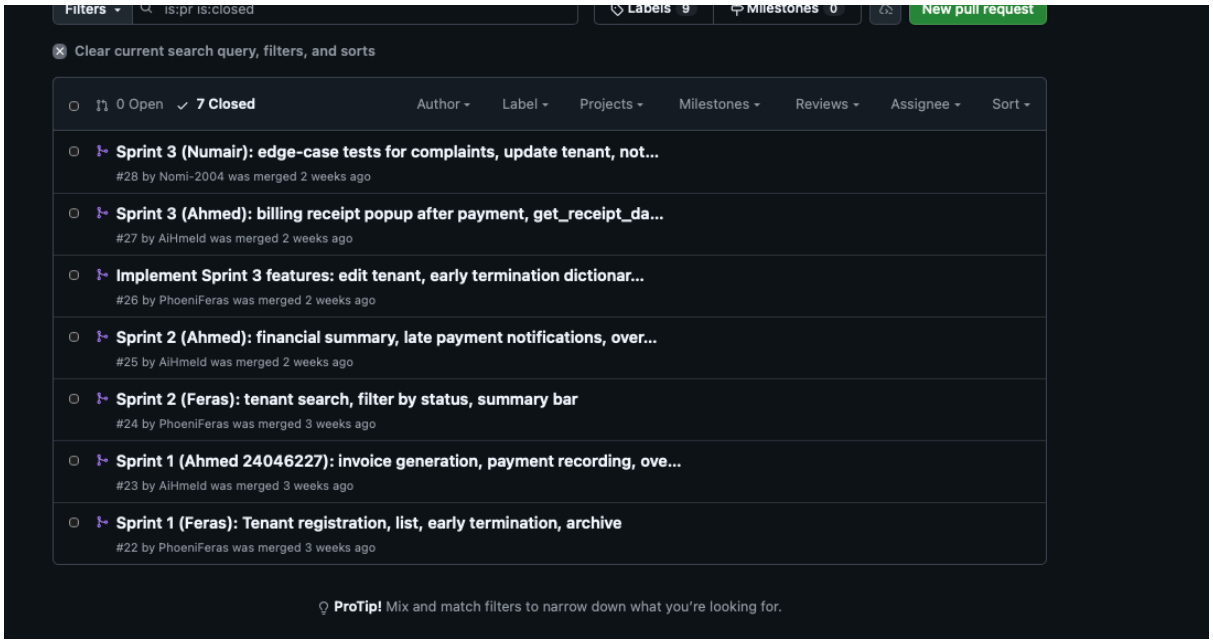
Sprint 4, Largest Sprint by Line Count

I designed and built the new visual layer from scratch. views/theme.py became the single source of truth for every colour, font, spacing value, and border radius. I added a full component library to views/ui_helpers.py — cards, KPI tiles, toast notifications, a confirm dialog, pill badges, empty states, segmented controls, search bars, and the notification dropdown. I then restyled every existing view on top of these primitives: login, main shell, dashboard, users, apartments, tenants, invoices, payments, maintenance, and reports. Every messagebox call was replaced with toast notifications, and a custom confirm_dialog was added that honours the dark theme. I wired up global keyboard shortcuts and replaced the Sprint 2 notification popup with an inline dropdown.

On the backend I added views/pdf_generator.py for fpdf2 invoice and receipt PDFs; database_manager._apply_migrations() to idempotently remove a legacy UNIQUE constraint; tenant role blocking at the login screen; invoice search by status and name; and CSV export on the reports module. Nine new test files brought automated coverage from 189 to 330 tests, all passing.

Submission deliverables I owned directly: writing README.txt and the root README.md; regenerating requirements.txt; drawing all UML diagrams in Astah, use case, class, and four sequence diagrams and exporting them to diagrams.pdf (23 pages, A3, committed 18 April); and running the final demo rehearsal with the full team.

```
2bedc1c Fix (Omar): universal root-level trackpad scroll — routes MouseWheel to nearest scrollable ancestor app-wide
0b9f4d0 Fix (Omar): trackpad scroll in CTkScrollableFrame — consistent with treeview scroll app-wide
ceddc72 Fix (Omar): wrap Cost Report tab in CTkScrollableFrame — Refresh button now reachable
758ecd4 Fix (Omar): Scheduled Date is optional in Assign section, clarify label
ea01d9b Fix (Omar): MNT-07 add Scheduled column to All Tickets, MNT-09 add * to date fields
9003c25 Fix (Omar): fluid tables — fill window width and redistribute proportionally on resize
9fda39c Fix (Omar): stretch=False + wider col padding — prevents autofit being overridden
5a48f8e Fix (Omar): autofit_columns + alternating row stripes across all tables app-wide
8a535cd Fix (Omar): dark-themed tables, auto-stretch columns, proper scrollbars — matches app design
ce3d935 Fix (Omar): Sprint 2 QA — role sidebar, table alignment, required field indicators, validation order, popup consistency
c4dd778 Docs (Omar): Sprint 2 master test plan — 80 manual tests across all modules
2f0dfde Sprint 2 (Numair): maintenance cost reports, date communication, tenant notification — merged to dev
920c8fe Tests (Omar): 40 unit tests for Numair Sprint 2 — cost reports, status counts, notify tenant
7c6309e Fix (Omar): wire get_monthly_revenue() to Financial Summary tab UI
38f754b Fix (Omar): guard archived_tree access in refresh_list to prevent crash on Tenants tab load
9c55ca0 Docs (Omar): Ahmed Sprint 2 merged — updated BACKLOG.md with completion status
62d5f3c Fix (Omar): patched Ahmed's invoice/payment views — added missing column headings, wrote 39 Sprint 2 unit tests
5cf464a Docs (Omar): remove config file from tracking
a4fbe30 Docs (Omar): clean up project guidelines wording
a40f931 Docs (Omar): set Numair's student ID (24026758), update project guidelines with full sprint 2 status
e5827ab Docs (Omar): fix Numair's sprint 2 guide — remove redundant import/heading loops, fix broken Notify Tenant layout instruction
4c53e05 Docs (Omar): fix Ahmed's sprint 2 guide — remove redundant imports/loops, fix method name mismatch, update progress tracker
5664be1 Docs (Omar): update project guidelines — brief alignment, team IDs, sprint status, submission checklist, missing requirements tracker
9f47973 Merge: integrate Feras Sprint 2 (search/filter) with Omar hotfixes (archived tab, assign guard, form reset)
dd007e5 Fix (Omar): patch Feras tenant module — student ID, terminated-tenant assign guard, date validation, form reset, archived tenants tab
f8e1613 Sprint 2 (Omar): Checked off completed reporting and notification foundation tasks
7b1defb Sprint 2 (Omar): Fixed critical crash bug caused by broken indentation in main_view.py
5aa73a4 Sprint 2 (Omar): Fixed notification badge UI packing layout
bd36b47 Sprint 2 (Omar): Added notification badge to main_view header
5a5ca7d Sprint 2 (Omar): 27 unit tests for ReportManager and NotificationManager
134ed96 Sprint 2 (Omar): Reports module (occupancy, financial, maintenance), notification manager, updated seed data
98fe6e7 Sprint 1 Complete: Reorganised guides into sprint1/ & sprint2/ folders, updated BACKLOG, added Sprint 2 guides for all members
8c7c006 Fix (Omar): comprehensive audit sweep — patched SQL injection, enforced business rules, hardened security, expanded tests to 27
f5fcfef UI (Omar): fixed massive frontend data cropping; tables now fully render Apartment IDs, Lease Start/End bounds, and Occupations to sync 1:1 with Backe
nd arrays
a0aa152 add time_spent and cost columns to all tickets table
e9be17f add 20 unit tests covering tenant, finance and maintenance managers
3fb3c28 fix logout not closing window, fix unhandled exceptions, fix cost overwriting, add missing confirms, fix table header sort
30e095a Fix (Omar): normalize macOS trackpad delta for Treeview scrolling to prevent freezing or hyperscrolling
6466642 Fix (Omar): removed useless CTkScrollableFrame wrapper from Tenant Register tab that caused truncation; replaced with standard CTkFrame
1194209 Fix (Omar): UI layout standardizations — all forms perfectly centered, labels right-aligned, entries left-aligned for consistent look
2636bf5 Fix (Omar): perfect horizontal alignment and styling for 3-column Manage tab layout
2e0f093 Fix (Omar): UX polish pass — enter key bind on login, trackpad/mousewheel scroll for all Treeviews via ui_helpers, and 3-column Manage tab layout to f
ix ugly padding
817e971 Fix (Omar): UI layout bugs — scrollable Manage tab, scrollbars on all Treeviews, Refresh buttons, full form resets, dead lease_period field removed
6382b5b Fix (Omar): 7 deep bugs — FK pragma, ghost apt_id, re-termination guard, dead-tenant invoicing, atomic payment tx, assign-apt UI, hollow staff role re
moved
e5fce21 Fix (Omar): existence validation across all managers + crash guard in maintenance_view
c010908 Fix (Omar): archive_tenant now raises error if Tenant ID does not exist
eedf7de Fix (Omar): Archive Tenant restricted to manager/admin only; manager gets Tenants tab back
(venv) → ASD-UWE-2026 git:(main) ✗
                                                                                                    ✓ CO4T: 20%  ✓ G3PH: 80%
```



7.2 Feras Bader (24033513), Tenant Management

I owned the tenant module across all four sprints.

Sprint 1

I wrote tenant_manager.py from scratch: register_tenant with bcrypt on the user row, assign_apartment flipping the apartment status to Occupied, archive_tenant, early_termination with the 5% penalty, and the list and filter helpers. I also wrote the first version of views/tenant_view.py with the Active Tenants tab, Register tab, and Assign

Apartment tab. The tenant columns, ni_number, full_name, apartment_requirements, and the lease fields came from reading the case study paragraph by paragraph to make sure nothing was missed.

Sprint 2

I added search_tenants(), filter_tenants_by_status(), and get_tenant_count_by_status() to the manager, wired them into a new search bar and status dropdown on the Active tab, and built the Archived/Terminated tab. The summary counts strip above the table came out of a Sprint 1 retrospective point, front-desk staff had no quick way to see how many tenants were in each status at a glance.

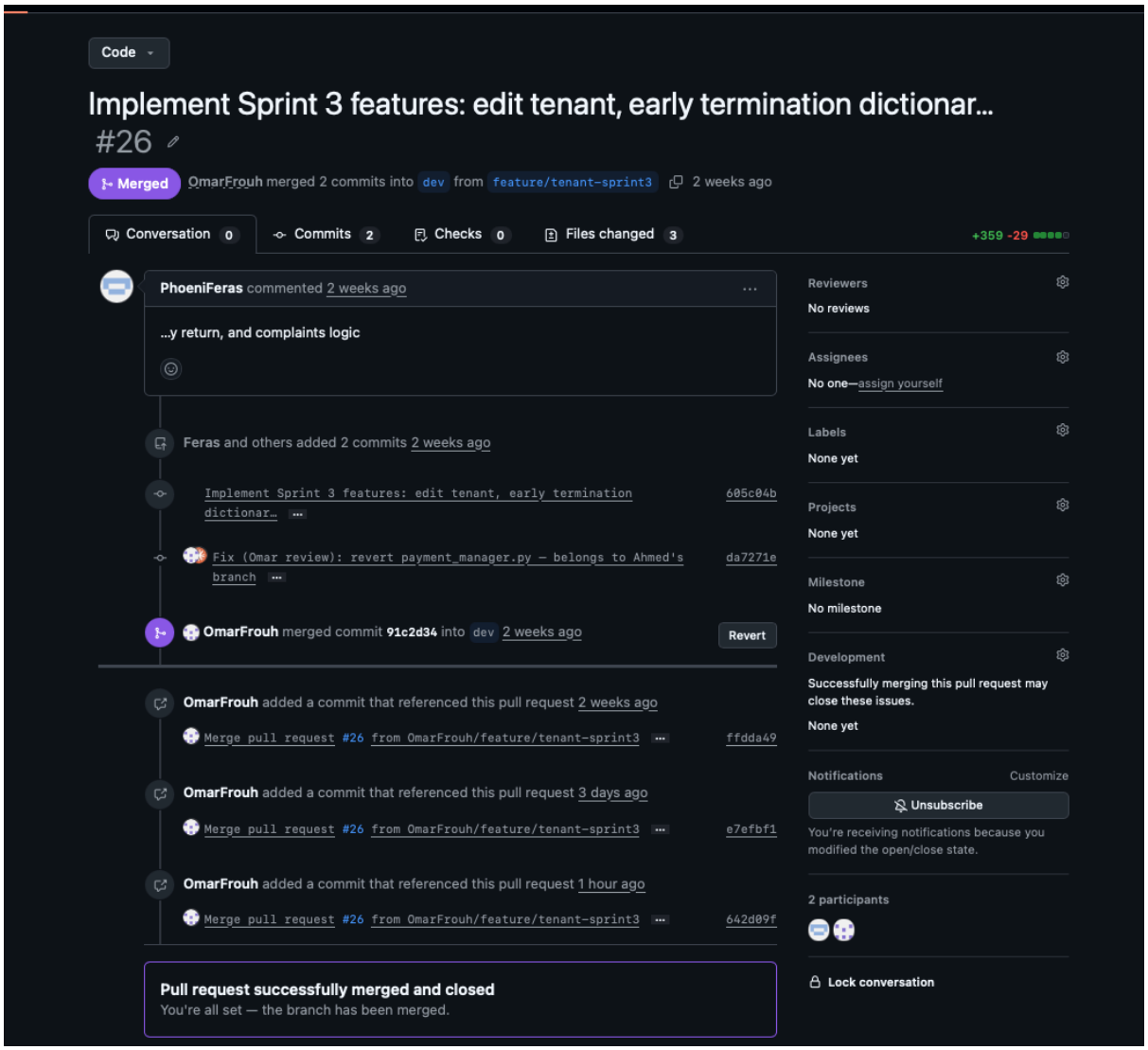
Sprint 3

I added the Edit Tenant tab to views/tenant_view.py, wrote get_tenant_by_id() and update_tenant() in the manager, built the Log Complaint and Resolve Complaint flows, and updated early_termination to enforce the one-month notice period, returning a dict with both the penalty amount and the effective date. This was delivered as PR #26 on 8 April.

Sprint 4

I worked with Omar on the tenant view restructure to make sure the seven tabs mapped sensibly onto the new card-based layout. I also own the Testing Document PDF (Element 3), which is submitted separately, I ran the full automated test suite to capture the pass evidence and compiled the manual test case table covering all four modules.

My tests sit in tests/test_tenant_manager.py and, from Sprint 3 onwards, in tests/test_sprint3.py. My student header (# Student Name: Feras Bader, # Student ID: 24033513) appears on tenant_manager.py and views/tenant_view.py.




```
test_update_terminated_tenant_fails (test_sprint3.TestUpdateTenant.test_update_terminated_tenant_fails) ... ok
test_archive_tenant_clears_apartment_ghost (test_tenant_manager.TestTenantManager.test_archive_tenant_clears_apartment_ghost) ... ok
test_assign_occupied_apartment_fails (test_tenant_manager.TestTenantManager.test_assign_occupied_apartment_fails) ... ok
test_early_termination_charges_penalty (test_tenant_manager.TestTenantManager.test_early_termination_charges_penalty) ... ok
test_register_and_assign_tenant (test_tenant_manager.TestTenantManager.test_register_and_assign_tenant) ... ok
test_reactivate_already_active_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_already_active_raises) ... ok
test_reactivate_archived_becomes_active (test_tenant_reactivate.TestTenantReactivate.test_reactivate_archived_becomes_active) ... ok
test_reactivate_nonexistent_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_nonexistent_raises) ... ok
test_reactivate_terminated_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_terminated_raises) ... ok

----------------------------------------------------------------------
Ran 330 tests in 75.027s

OK
(venv) → ASD-UWE-2026 git:(main) ✗
```

```
(venv) → ASD-UWE-2026 git:(main) ✗ git log --oneline --author="Feras" | head -100
e848397 Implement Sprint 3 features: edit tenant, early termination dictionary return, and complaints logic
dc149e1 Merge pull request #24 from OmarFrouh/feature/tenant-sprint2
bcb1a14 Sprint 2 (Feras): tenant search, filter by status, summary bar
d26282e Merge pull request #22 from OmarFrouh/feature/tenant-management
e2b9016 Sprint 1 (Feras): Tenant registration, list, early termination, archive
(venv) → ASD-UWE-2026 git:(main) ✗
```

7.3 Ahmed Deraz (24046227) — Payment and Billing

I was responsible for the finance side of the application: invoices, payments, overdue tracking, and the receipt workflow.

Sprint 1

I wrote invoice_manager.py and payment_manager.py. The most important detail in record_payment was atomicity, recording a payment involves two related writes (an INSERT into payments, then an UPDATE to invoices.status when the balance is cleared) and both have to succeed or roll back together. generate_invoice reads monthly_rent from the tenant row and creates an invoice with a 14-day due date. On the UI side I built views/invoice_view.py and views/payment_view.py with the Generate, List, Overdue, and Record Payment tabs.

Sprint 2

I added get_overdue_with_details() and get_invoice_aging() to the invoice manager, and get_collected_vs_pending(), get_payment_history_by_method(), and get_monthly_revenue() to the payment manager. On the UI side I added the overdue badge on the invoice list, the Late Payment Notifications tab, and the Financial Summary tab with the payment method breakdown table and the six-month revenue chart. I wrote 39 unit tests in tests/test_ahmed_sprint2.py.

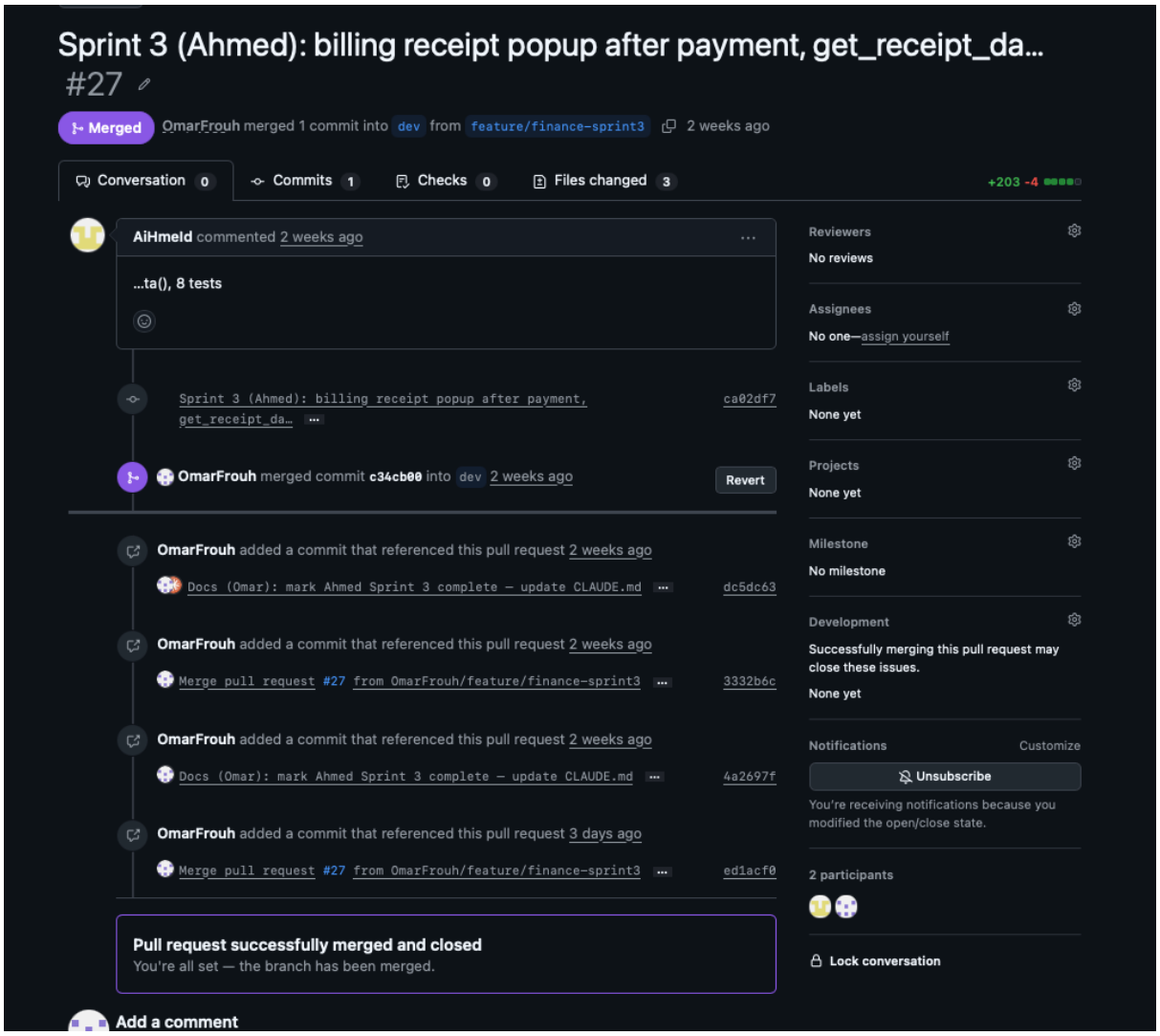
Sprint 3

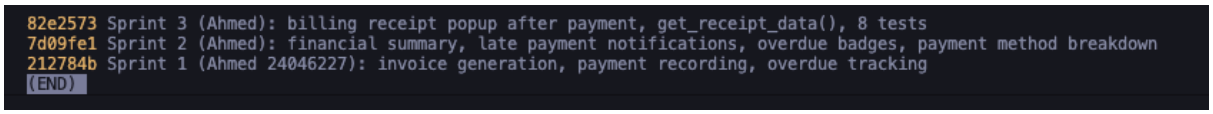
I addressed the brief requirement for billing receipts. I added get_receipt_data() to PaymentManager, joining payment, invoice, tenant, and apartment rows into a single dictionary, and modified record_payment to return the new payment_id so the view could fetch the receipt immediately after recording. I then built the receipt popup as a CTkToplevel window and wrote eight additional tests. This was delivered as PR #27 on 9 April.

Sprint 4

I kept the receipt popup but rebuilt it on top of Omar's new card primitives so it matches the dark theme. The PDF export was wired in so that after a payment is recorded, the receipt popup offers a Save PDF button that writes through views/pdf_generator.py:generate_receipt_pdf. I also took part in the QA run on the finance flows before the Sprint 4 merge.

My tests are in tests/test_ahmed_sprint2.py (39 Sprint 2 tests) and the Sprint 3 receipt tests in tests/test_sprint3.py. My student header (# Student Name: Ahmed Deraz, # Student ID: 24046227) appears on the relevant files.





7.4 Numair (24026758) — Maintenance and Testing

I owned the maintenance component and, from Sprint 3, the edge-case testing work across the whole application.

Sprint 1

I wrote maintenance_manager.py with create_ticket, assign_staff, update_status, log_costs, and the list and filter helpers. I built the first version of views/maintenance_view.py with the New Ticket tab, All Tickets tab, and Manage tab (updating status, assigning staff, logging costs). The priority and status vocabularies, Low/Medium/High/Urgent and Open/Assigned/In Progress/Fixed/Closed came directly from the brief's description of the maintenance

workflow.

Sprint 2

I added the reporting side: get_cost_report_by_apartment() and get_cost_report_by_city() on the manager, and a new Cost Report tab with KPI summary cards and per-apartment and per-city tables. I also added the Notify Tenant section to the Manage tab, which writes a scheduled maintenance date onto the ticket for the tenant view to surface. I wrote 40 unit tests in tests/test_numair_sprint2.py, and added get_tickets_by_status_count() to feed the Sprint 4 dashboard KPI.

Sprint 3

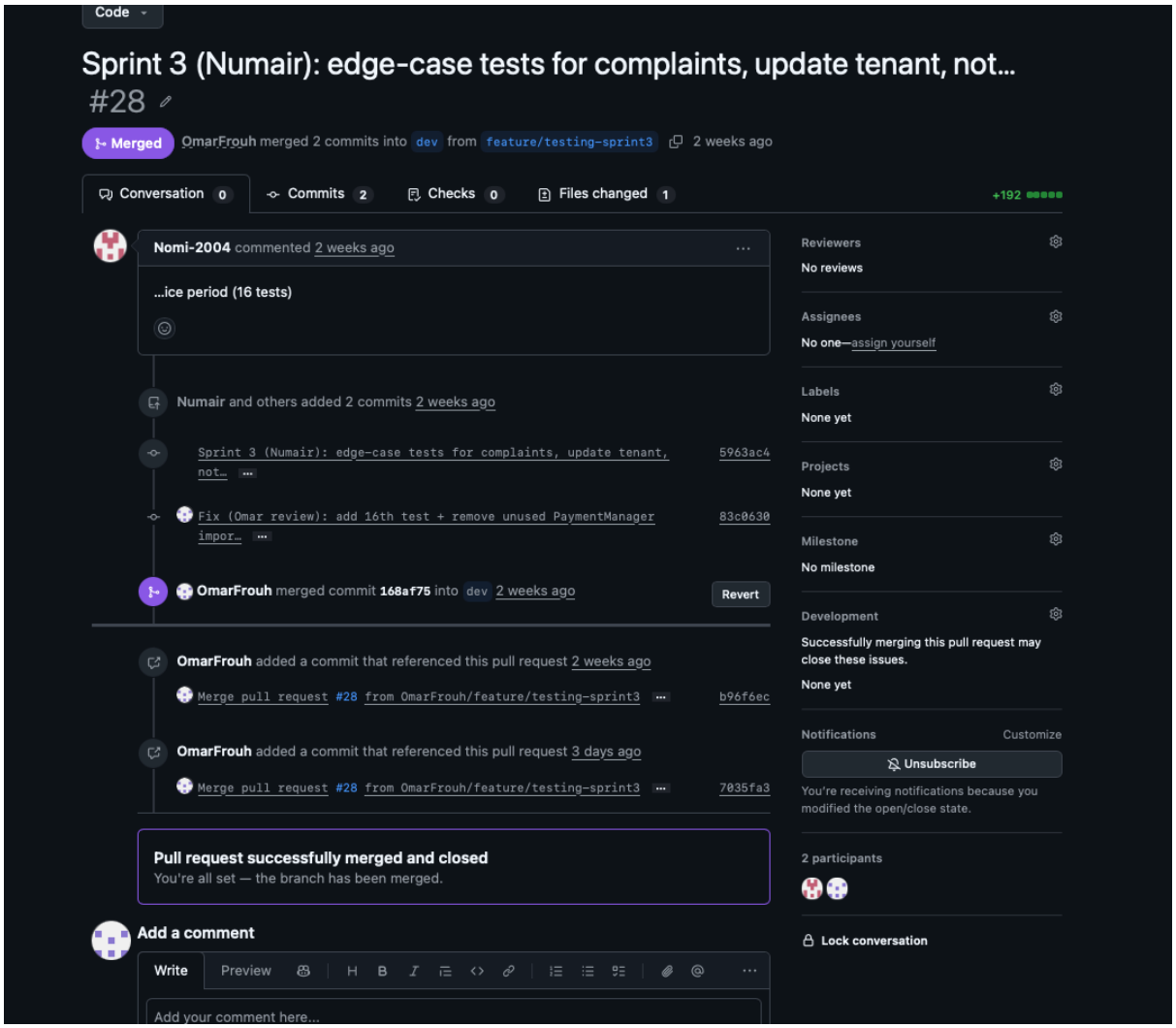
My role broadened this sprint. On top of maintenance fixes, I wrote 16 edge-case tests in tests/test_numair_sprint3.py specifically designed to stress the system: logging a complaint for one tenant then verifying a second tenant cannot see it; partial tenant updates where only one field changes and the rest stay untouched; and notice period boundary dates — today, today plus exactly 30 days, and the day before. PR #28 was delivered on 12 April.

Sprint 4

I pair-debugged a notification dropdown edge case with Omar during the redesign merge, and did a full manual pass through the Maintenance view to confirm none of the ticket workflows were broken by the UI changes.

My tests are in tests/test_numair_sprint2.py (40 Sprint 2 tests) and tests/test_numair_sprint3.py (16 Sprint 3 edge-case tests). My student header (# Student Name: Numair, # Student ID: 24026758) appears on the relevant files.

```
(venv) → ASD-UWE-2026 git:(main) ✗ git log --oneline --author="Numair" | head -100
d98d680 Sprint 3 (Numair): edge-case tests for complaints, update tenant, notice period (16 tests)
88c8dc2 Sprint 2 (Numair): maintenance cost report (by apartment & city), date communication, tenant notification
1bad68a Sprint 1 (Numair): Maintenance tickets, staff assignment, cost logging, status updates
(venv) → ASD-UWE-2026 git:(main) ✗
```



8. Retrospective and Lessons Learned

8.1 What Went Well

Sprint-by-sprint delivery prevented a last-week panic. Because dev was always runnable at the end of each sprint, there was never a decision in the final 48 hours about whether to cut a feature. By the time Sprint 4 opened, the only outstanding items were polish and paperwork.

Single reviewer worked at this team size. Routing every pull request through Omar was a theoretical bottleneck but in practice meant one person understood what was entering dev and could catch cross-cutting bugs early. This approach would not scale to a team of eight.

Automated tests enabled a confident UI redesign. With 189 tests passing before Sprint 4, the full UI redesign in ten days was feasible without fear of breaking invisible behaviour. When a Sprint 4 refactor did fail a test, it was usually caught within minutes.

8.2 What Could Have Gone Better

The brief was read too quickly initially. Four requirements (editing tenants, one-month notice, complaint logs, billing receipts) were only identified in the Sprint 2 retrospective. A line-by-line brief review during Sprint 1 planning would have been more thorough. The time saved by skipping it cost a full sprint later.

Async standups were low signal. WhatsApp updates were better than nothing, but a ten-minute voice call at the start of each lab day would have surfaced integration clashes earlier than the PR review stage.

UI polish was deferred too long. The functional application existed by end of Sprint 2, but visual consistency work was concentrated into Sprint 4. A small design-sweep slot in every sprint from Sprint 1 would have distributed this workload more evenly.

8.3 What the Team Would Do Differently

7. Conduct a line-by-line brief check at Sprint 1 planning, verified against every sprint's backlog.
8. Add a lightweight daily voice standup on lab days (10 minutes, not the full 15).
9. Schedule a 20-minute design sweep slot at the end of every sprint from Sprint 1, so Sprint 4 does not absorb all UI work.

8.4 How Agile Actually Helped

Agile allowed the team to change the plan without experiencing the change as a failure. Without Scrum, discovering four missing requirements in week four would have felt like a crisis and might have resulted in cutting features or rushed implementation. Within the Scrum framework, it became a retrospective item that turned directly into the next sprint. Every feature that ended up in PAMS is present in the brief, and that outcome was achieved by iterating rather than by getting the plan right first time.

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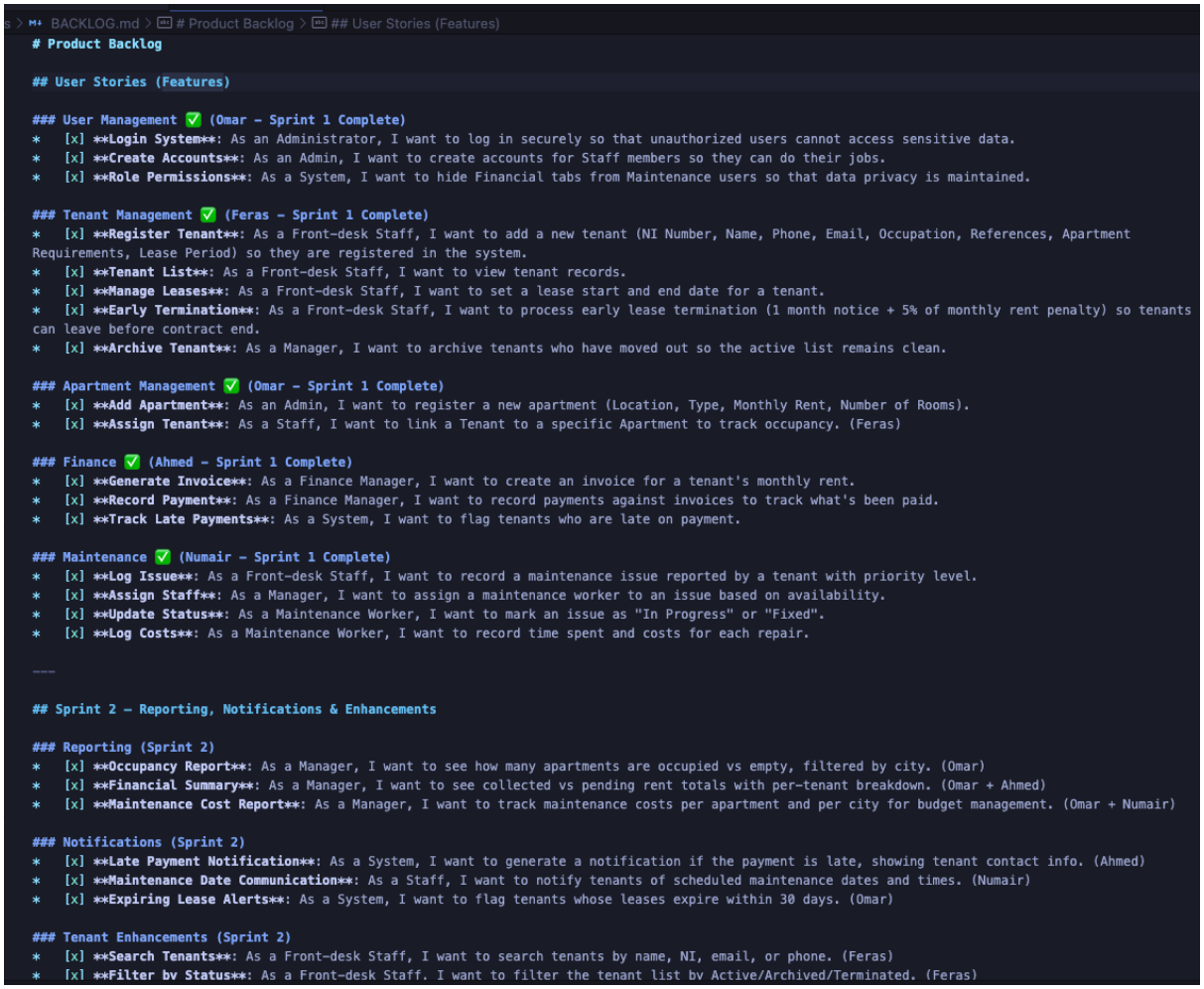
9. Conclusion

PAMS covers every core component listed in the brief: account and user management (five roles, bcrypt, role-based sidebar); tenant management (registration, update, lease, complaint log, early termination with one-month notice and 5% penalty); apartment management (register, assign, lifecycle); payment and billing (invoice, payment with atomic transactions, receipts, overdue tracking and notifications); reporting (occupancy, financial, maintenance cost, all filterable by city); and maintenance (tickets, priority, staff assignment, cost logging, tenant notification). All of this is built on parameterised SQL, bcrypt password hashing, and 330 automated tests.

On the process side, the team ran four two-week sprints, used GitHub pull requests and a project board, held retrospectives that produced concrete changes to the next sprint's plan, and generated per-member contribution evidence supported by both this report and the git history.

10. Appendices

Appendix A — Project Backlog



BACKLOG.md > # Product Backlog > ## User Stories (Features)

```
# Product Backlog

## User Stories (Features)

### User Management ✅ (Omar – Sprint 1 Complete)
*   [x] **Login System**: As an Administrator, I want to log in securely so that unauthorized users cannot access sensitive data.
*   [x] **Create Accounts**: As an Admin, I want to create accounts for Staff members so they can do their jobs.
*   [x] **Role Permissions**: As a System, I want to hide Financial tabs from Maintenance users so that data privacy is maintained.

### Tenant Management ✅ (Feras – Sprint 1 Complete)
*   [x] **Register Tenant**: As a Front-desk Staff, I want to add a new tenant (NI Number, Name, Phone, Email, Occupation, References, Apartment
Requirements, Lease Period) so they are registered in the system.
*   [x] **Tenant List**: As a Front-desk Staff, I want to view tenant records.
*   [x] **Manage Leases**: As a Front-desk Staff, I want to set a lease start and end date for a tenant.
*   [x] **Early Termination**: As a Front-desk Staff, I want to process early lease termination (1 month notice + 5% of monthly rent penalty) so tenants
can leave before contract end.
*   [x] **Archive Tenant**: As a Manager, I want to archive tenants who have moved out so the active list remains clean.

### Apartment Management ✅ (Omar – Sprint 1 Complete)
*   [x] **Add Apartment**: As an Admin, I want to register a new apartment (Location, Type, Monthly Rent, Number of Rooms).
*   [x] **Assign Tenant**: As a Staff, I want to link a Tenant to a specific Apartment to track occupancy. (Feras)

### Finance ✅ (Ahmed – Sprint 1 Complete)
*   [x] **Generate Invoice**: As a Finance Manager, I want to create an invoice for a tenant's monthly rent.
*   [x] **Record Payment**: As a Finance Manager, I want to record payments against invoices to track what's been paid.
*   [x] **Track Late Payments**: As a System, I want to flag tenants who are late on payment.

### Maintenance ✅ (Numair – Sprint 1 Complete)
*   [x] **Log Issue**: As a Front-desk Staff, I want to record a maintenance issue reported by a tenant with priority level.
*   [x] **Assign Staff**: As a Manager, I want to assign a maintenance worker to an issue based on availability.
*   [x] **Update Status**: As a Maintenance Worker, I want to mark an issue as "In Progress" or "Fixed".
*   [x] **Log Costs**: As a Maintenance Worker, I want to record time spent and costs for each repair.

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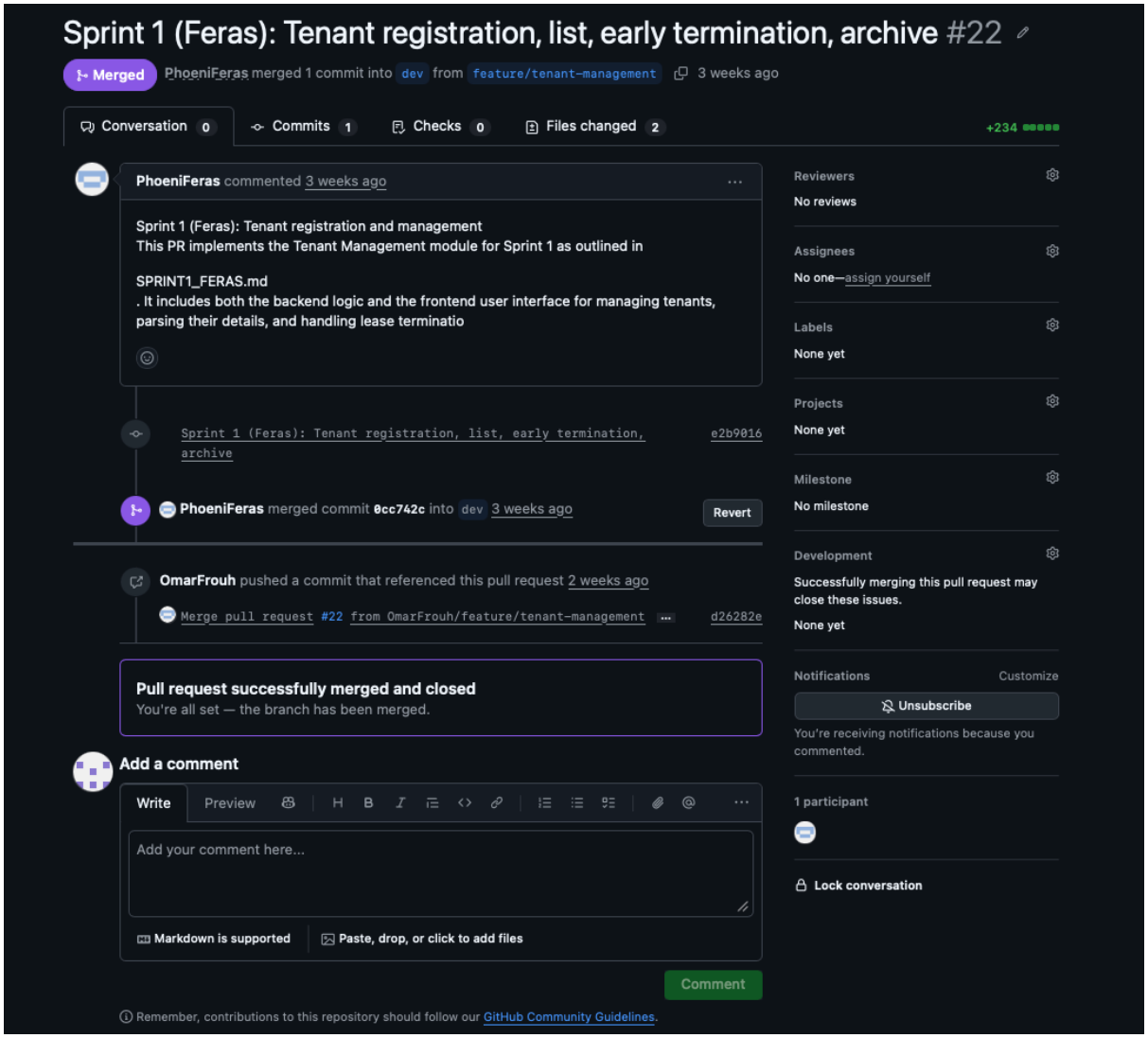
## Sprint 2 – Reporting, Notifications & Enhancements

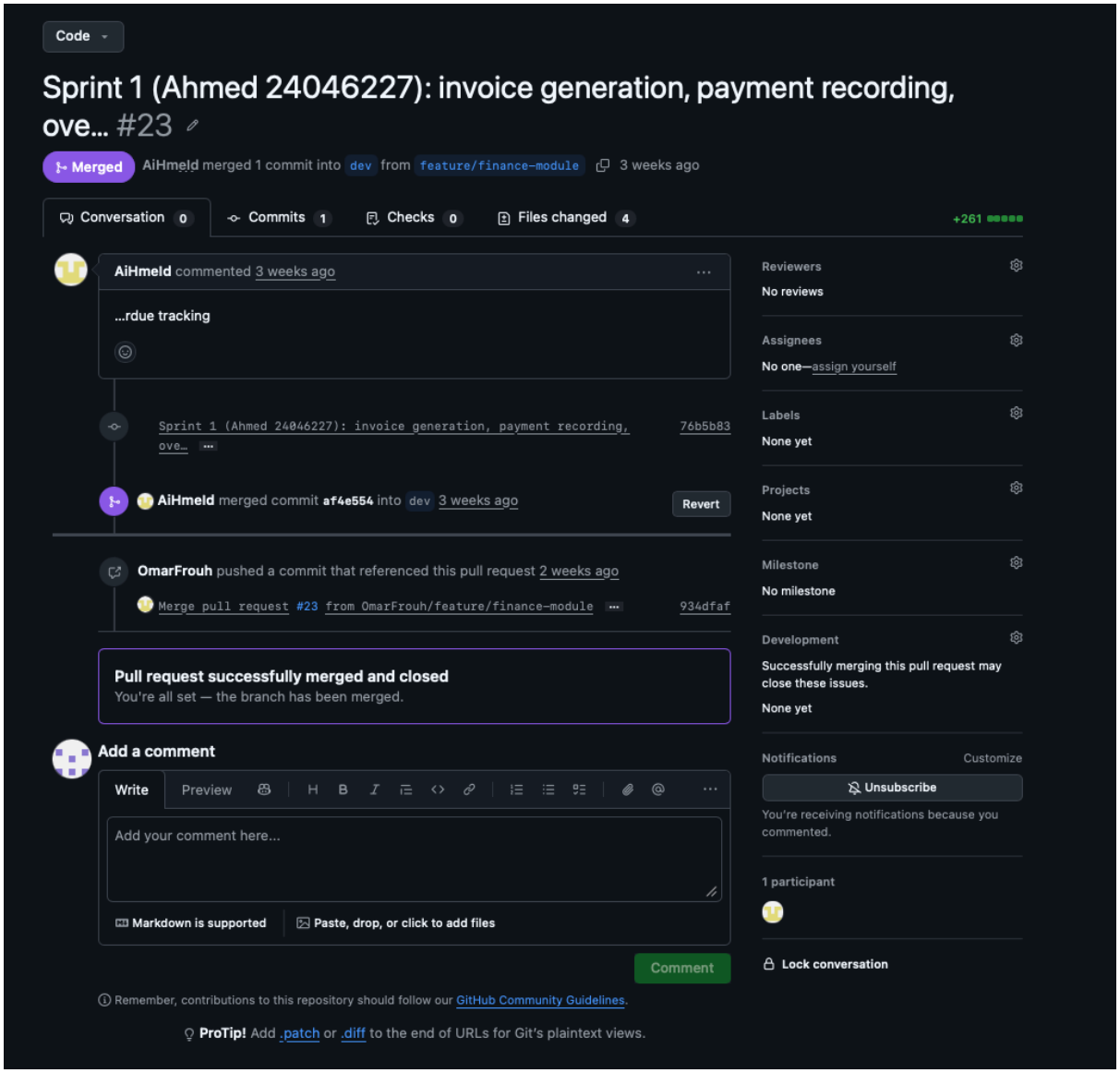
### Reporting (Sprint 2)
*   [x] **Occupancy Report**: As a Manager, I want to see how many apartments are occupied vs empty, filtered by city. (Omar)
*   [x] **Financial Summary**: As a Manager, I want to see collected vs pending rent totals with per-tenant breakdown. (Omar + Ahmed)
*   [x] **Maintenance Cost Report**: As a Manager, I want to track maintenance costs per apartment and per city for budget management. (Omar + Numair)

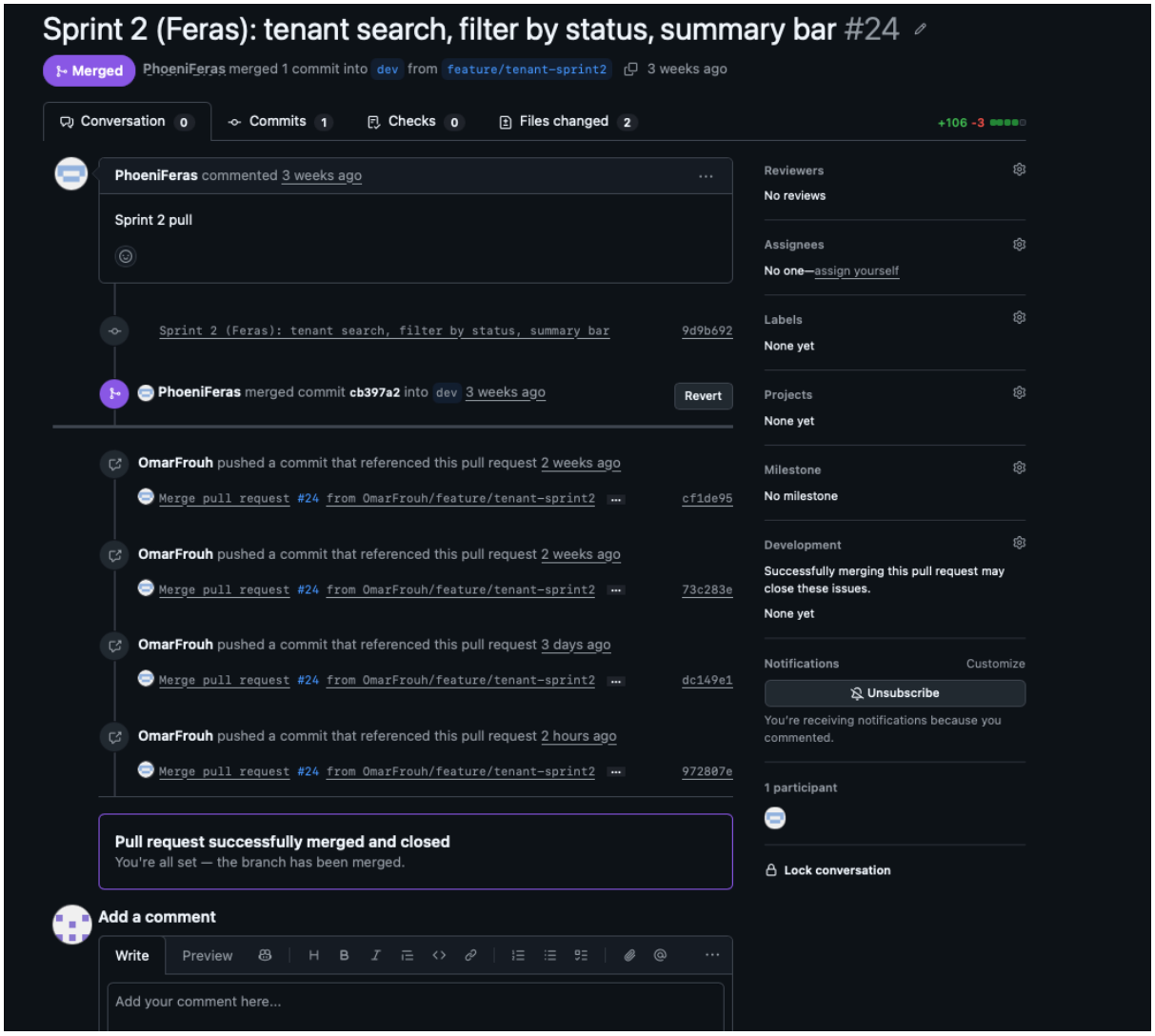
### Notifications (Sprint 2)
*   [x] **Late Payment Notification**: As a System, I want to generate a notification if the payment is late, showing tenant contact info. (Ahmed)
*   [x] **Maintenance Date Communication**: As a Staff, I want to notify tenants of scheduled maintenance dates and times. (Numair)
*   [x] **Expiring Lease Alerts**: As a System, I want to flag tenants whose leases expire within 30 days. (Omar)

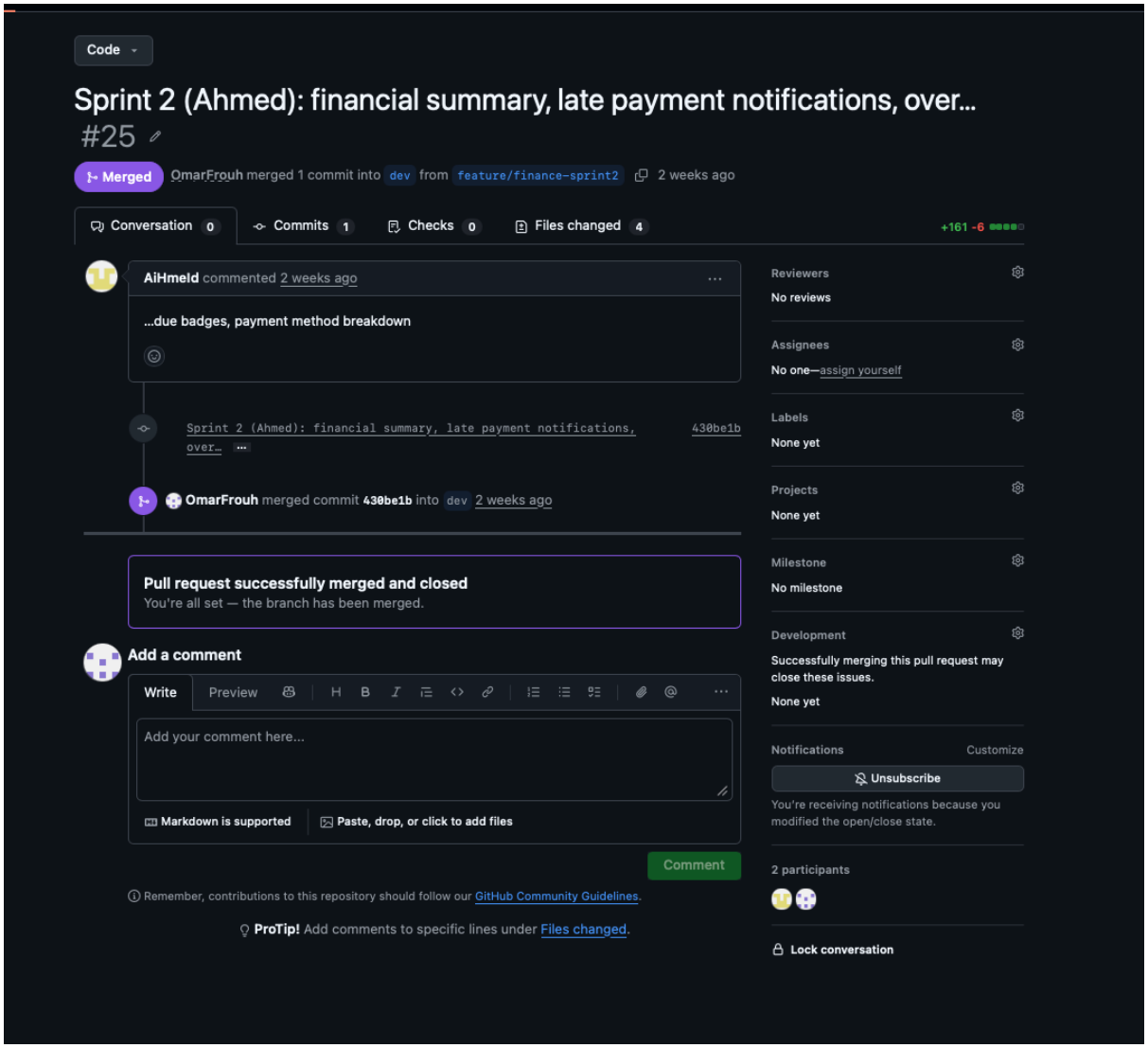
### Tenant Enhancements (Sprint 2)
*   [x] **Search Tenants**: As a Front-desk Staff, I want to search tenants by name, NI, email, or phone. (Feras)
*   [x] **Filter by Status**: As a Front-desk Staff, I want to filter the tenant list by Active/Archived/Terminated. (Feras)
```

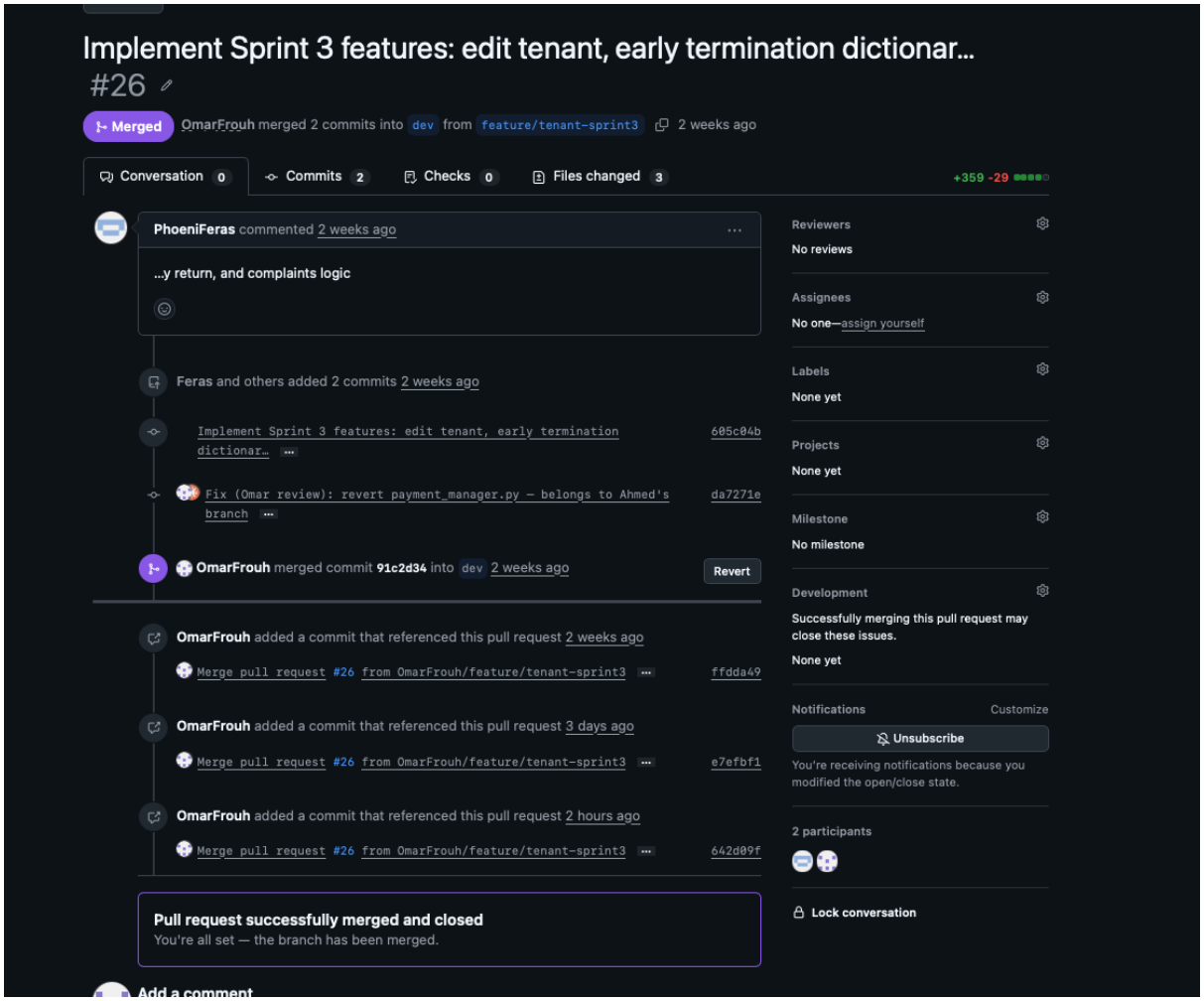
Appendix B — Pull Requests

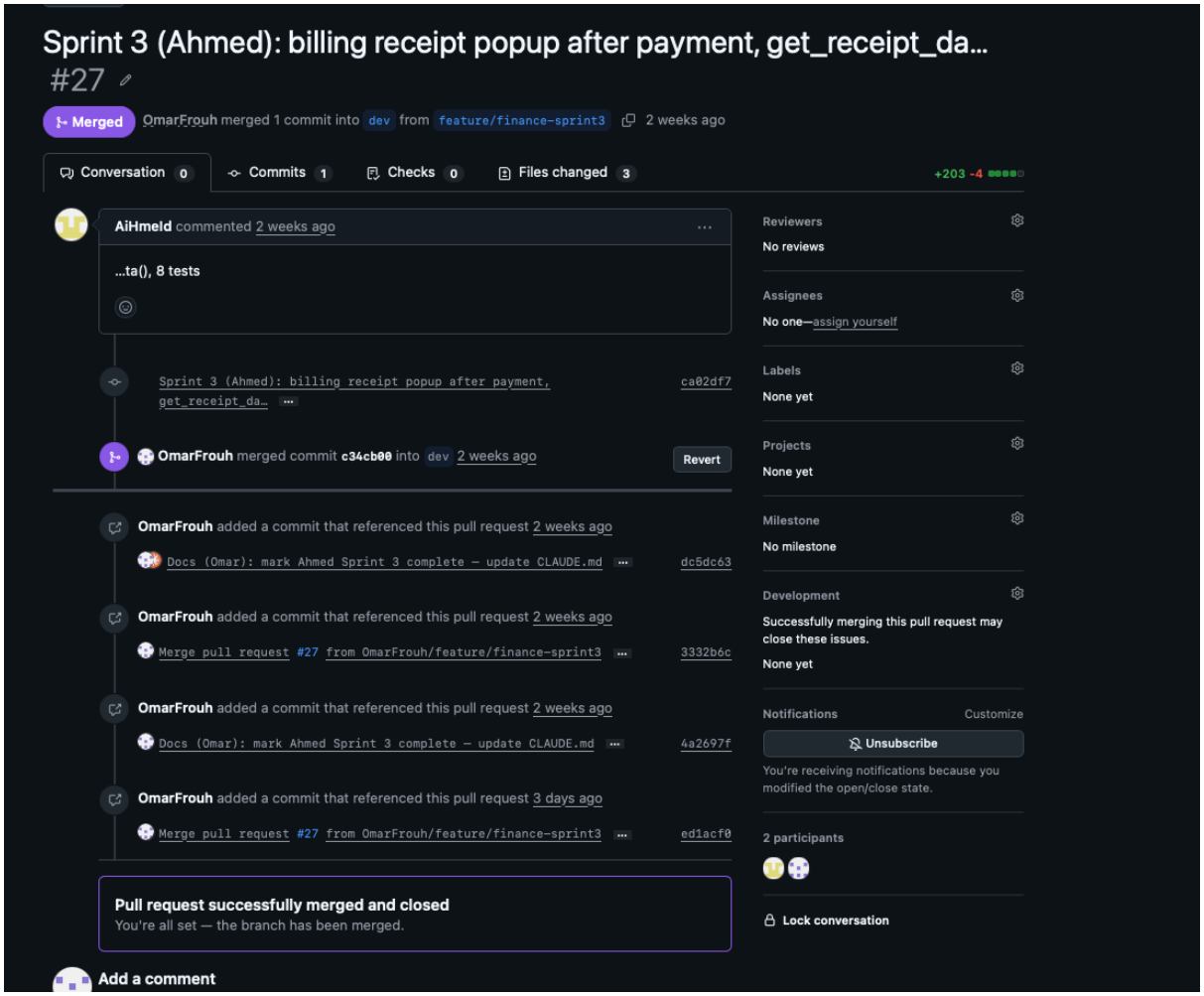


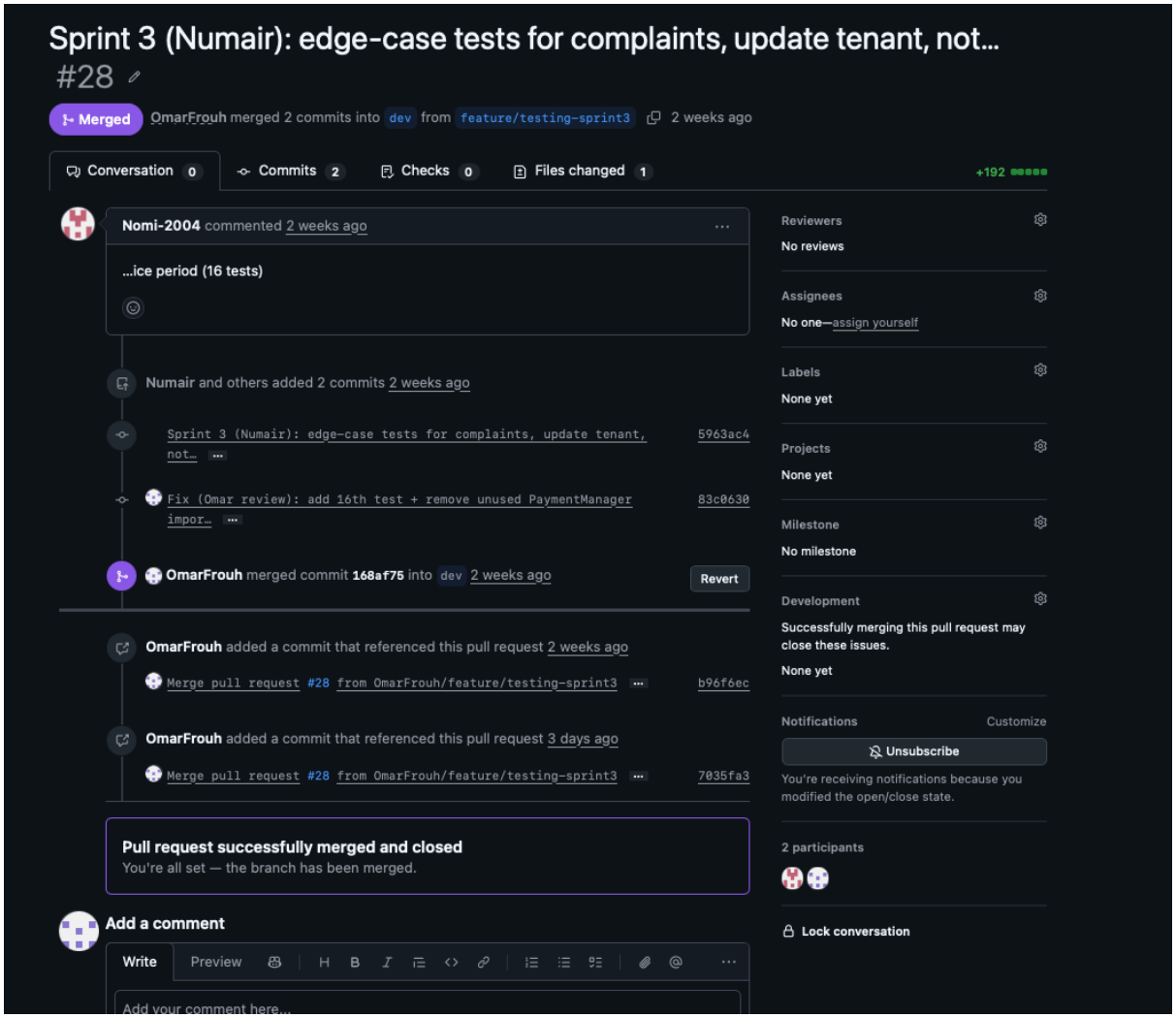

Code
Sprint 1 (Ahmed 24046227): invoice generation, payment recording, ove... #23
Merged
AiHmeld merged 1 commit into dev from feature/finance-module 3 weeks ago
Conversation 0
Commits 1
Checks 0
Files changed 4
+261
AiHmeld commented 3 weeks ago
...rdue tracking
Sprint 1 (Ahmed 24046227): invoice generation, payment recording, ove... 76b5b83
AiHmeld merged commit af4e554 into dev 3 weeks ago
Revert
OmarFrouh pushed a commit that referenced this pull request 2 weeks ago
Merge pull request #23 from OmarFrouh/feature/finance-module 934dfaf
Pull request successfully merged and closed
You're all set — the branch has been merged.
Add a comment
Write
Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Comment
Remember, contributions to this repository should follow our GitHub Community Guidelines.
ProTip! Add .patch or .diff to the end of URLs for Git's plaintext views.
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you commented.
1 participant
Lock conversation

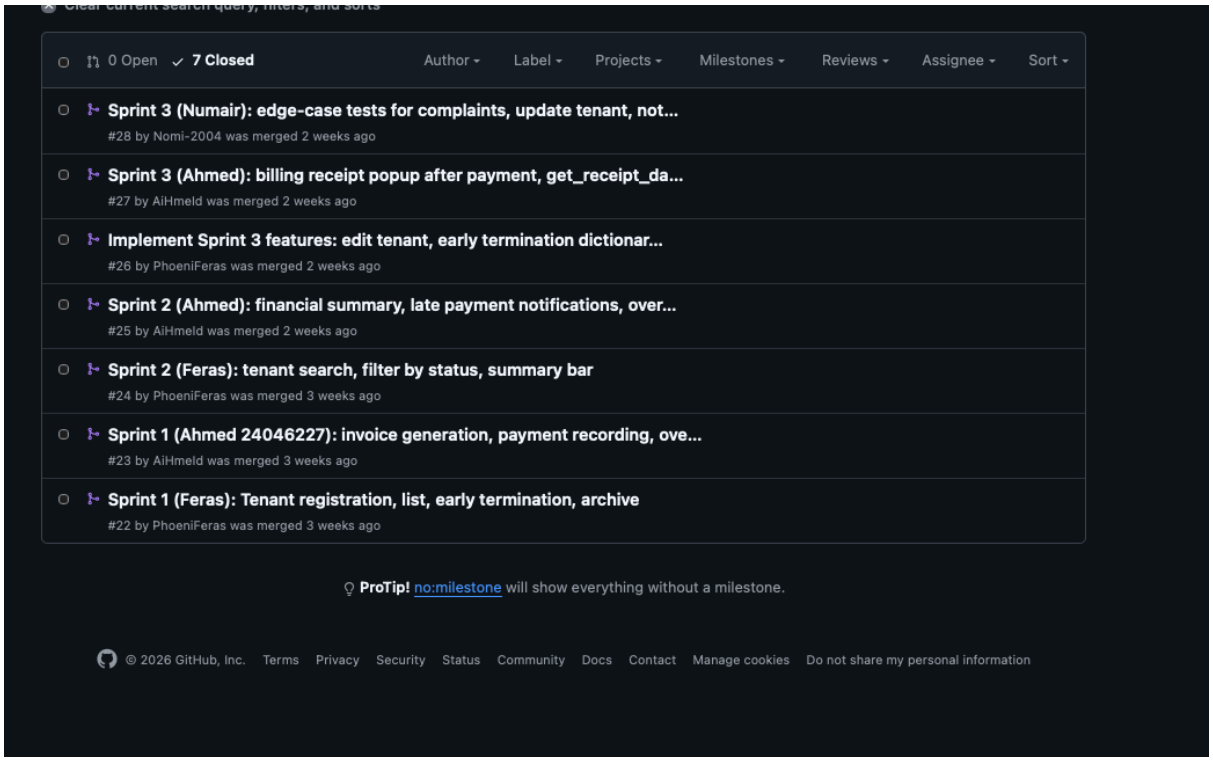

Sprint 2 (Feras): tenant search, filter by status, summary bar #24
Merged
PhoeniFeras merged 1 commit into dev from feature/tenant-sprint2
3 weeks ago
Conversation 0
Commits 1
Checks 0
Files changed 2
+106 -3
PhoeniFeras commented 3 weeks ago
Sprint 2 pull
Sprint 2 (Feras): tenant search, filter by status, summary bar 9d9b692
PhoeniFeras merged commit cb397a2 into dev 3 weeks ago
Revert
OmarFrouh pushed a commit that referenced this pull request 2 weeks ago
Merge pull request #24 from OmarFrouh/feature/tenant-sprint2 cf1de95
OmarFrouh pushed a commit that referenced this pull request 2 weeks ago
Merge pull request #24 from OmarFrouh/feature/tenant-sprint2 73c283e
OmarFrouh pushed a commit that referenced this pull request 3 days ago
Merge pull request #24 from OmarFrouh/feature/tenant-sprint2 dc149e1
OmarFrouh pushed a commit that referenced this pull request 2 hours ago
Merge pull request #24 from OmarFrouh/feature/tenant-sprint2 972807e
Pull request successfully merged and closed
You're all set — the branch has been merged.
Add a comment
Write
Preview
Add your comment here...
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you commented.
1 participant
Lock conversation


Code
Sprint 2 (Ahmed): financial summary, late payment notifications, over...
#25
Merged
OmarFrouh merged 1 commit into dev from feature/finance-sprint2 2 weeks ago
Conversation 0
Commits 1
Checks 0
Files changed 4
+161 -6
AiHmeId commented 2 weeks ago
...due badges, payment method breakdown
Sprint 2 (Ahmed): financial summary, late payment notifications, over... 430be1b
OmarFrouh merged commit 430be1b into dev 2 weeks ago
Pull request successfully merged and closed
You're all set — the branch has been merged.
Add a comment
Write
Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Comment
Remember, contributions to this repository should follow our GitHub Community Guidelines.
ProTip! Add comments to specific lines under Files changed.
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.
2 participants
Lock conversation


Implement Sprint 3 features: edit tenant, early termination dictionar...
#26
Merged
OmarFrouh merged 2 commits into dev from feature/tenant-sprint3 2 weeks ago
Conversation 0
Commits 2
Checks 0
Files changed 3
+359 -29
PhoeniFeras commented 2 weeks ago
...y return, and complaints logic
Feras and others added 2 commits 2 weeks ago
Implement Sprint 3 features: edit tenant, early termination dictionar… 605c04b
Fix (Omar review): revert payment_manager.py — belongs to Ahmed's branch da7271e
OmarFrouh merged commit 91c2d34 into dev 2 weeks ago
Revert
OmarFrouh added a commit that referenced this pull request 2 weeks ago
Merge pull request #26 from OmarFrouh/feature/tenant-sprint3 ffdda49
OmarFrouh added a commit that referenced this pull request 3 days ago
Merge pull request #26 from OmarFrouh/feature/tenant-sprint3 e7efbf1
OmarFrouh added a commit that referenced this pull request 2 hours ago
Merge pull request #26 from OmarFrouh/feature/tenant-sprint3 642d09f
Pull request successfully merged and closed
You're all set — the branch has been merged.
Add a comment
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.
2 participants
Lock conversation


Sprint 3 (Ahmed): billing receipt popup after payment, get_receipt_da...
#27
Merged
OmarFrouh merged 1 commit into dev from feature/finance-sprint3 2 weeks ago
Conversation 0
Commits 1
Checks 0
Files changed 3
+203 -4
AiHmeld commented 2 weeks ago
...ta(), 8 tests
Sprint 3 (Ahmed): billing receipt popup after payment, get_receipt_da… ca02df7
OmarFrouh merged commit c34cb00 into dev 2 weeks ago
Revert
OmarFrouh added a commit that referenced this pull request 2 weeks ago
Docs (Omar): mark Ahmed Sprint 3 complete — update CLAUDE.md dc5dc63
OmarFrouh added a commit that referenced this pull request 2 weeks ago
Merge pull request #27 from OmarFrouh/feature/finance-sprint3 3332b6c
OmarFrouh added a commit that referenced this pull request 2 weeks ago
Docs (Omar): mark Ahmed Sprint 3 complete — update CLAUDE.md 4a2697f
OmarFrouh added a commit that referenced this pull request 3 days ago
Merge pull request #27 from OmarFrouh/feature/finance-sprint3 ed1acf0
Pull request successfully merged and closed
You're all set — the branch has been merged.
Add a comment
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.
2 participants
Lock conversation


Sprint 3 (Numair): edge-case tests for complaints, update tenant, not...
#28
Merged
OmarFrouh merged 2 commits into dev from feature/testing-sprint3 2 weeks ago
Conversation 0
Commits 2
Checks 0
Files changed 1
+192
Nomi-2004 commented 2 weeks ago
...ice period (16 tests)
Numair and others added 2 commits 2 weeks ago
Sprint 3 (Numair): edge-case tests for complaints, update tenant, not… 5963ac4
Fix (Omar review): add 16th test + remove unused PaymentManager impor… 83c0630
OmarFrouh merged commit 168af75 into dev 2 weeks ago
Revert
OmarFrouh added a commit that referenced this pull request 2 weeks ago
Merge pull request #28 from OmarFrouh/feature/testing-sprint3 b96f6ec
OmarFrouh added a commit that referenced this pull request 3 days ago
Merge pull request #28 from OmarFrouh/feature/testing-sprint3 7035fa3
Pull request successfully merged and closed
You're all set — the branch has been merged.
Add a comment
Write
Preview
Add your comment here...
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.
2 participants
Lock conversation


0 Open
7 Closed
Author
Label
Projects
Milestones
Reviews
Assignee
Sort
Sprint 3 (Numair): edge-case tests for complaints, update tenant, not...
#28 by Nomi-2004 was merged 2 weeks ago
Sprint 3 (Ahmed): billing receipt popup after payment, get_receipt_da...
#27 by AiHmeld was merged 2 weeks ago
Implement Sprint 3 features: edit tenant, early termination dictionar...
#26 by PhoeniFeras was merged 2 weeks ago
Sprint 2 (Ahmed): financial summary, late payment notifications, over...
#25 by AiHmeld was merged 2 weeks ago
Sprint 2 (Feras): tenant search, filter by status, summary bar
#24 by PhoeniFeras was merged 3 weeks ago
Sprint 1 (Ahmed 24046227): invoice generation, payment recording, ove...
#23 by AiHmeld was merged 3 weeks ago
Sprint 1 (Feras): Tenant registration, list, early termination, archive
#22 by PhoeniFeras was merged 3 weeks ago
ProTip! no:milestone will show everything without a milestone.
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Appendix C — Branches and Git Log


Branch
main
dev
feature/finance-module
feature/testing-sprint3
feature/tenant-sprint3
feature/tenant-sprint2
feature/tenant-management
feature/sprint4-submission
feature/sprint4-redesign
feature/sprint1-omar
feature/maintenance-sprint2
feature/maintenance-module
feature/finance-sprint3
feature/finance-sprint2

```
(venv) → ASD-UWE-2026 git:(main) ✗ git log --oneline --graph --all | head -60
* 719a2f1 update readme docs links
* 6bba6d5 remove mailmap
* d7d0fce add mailmap
* 010a5c5 tidy up backlog doc
* 309698a remove gitignore from tracking, move internal docs to archive
* 169d173 restore gitignore, add diagrams pdf
*   d98f054 merge dev into main: testing document and final fixes
|\
| * b3585a6 add testing document (element 3)
| * 89f2a67 clean up unused code and outdated docs
| * 9b529d8 fix invoice pdf export crash on non-latin-1 characters in tenant fields
* | 2ad8d56 merge cleanup and doc refresh into main
|\|
| * 711c9d5 clean up repo and refresh planning docs
* | 606ffeb merge dev into main: sprint 4 final submission
|\|
| * dc1573b update readme files: test count 189 to 330, add fpdf2 dependency
| * 21b855e move planning docs to docs/ folder
| * 8107f1e remove local dev notes from repo
| * 251ccc2 add split astah project under diagrams/ and remove stale pmas.asta
| *   ebe6a2d merge sprint 4 redesign into dev
| |\
| | * d86d206 update views with sprint 4 fixes; block tenant login; rename test data
| | * 6d925a4 add sprint 4 backend methods, db migration, extended tests, and pdf generator
| | * ab03111 wrap all tabviews in bordered card shell for visual consistency
| | * 6f9e055 fix pack/grid conflict in register tab: use grid-compatible section headers
| | * 8f29dd6 fix form resizability: sticky=ew, weight=1 columns, no fixed entry widths
| | * 8603e8c notification dropdown, keyboard shortcuts, empty states, hover highlight
| | * f6dbcbf redesign tenants view
| | * 9187c8b redesign maintenance view
| | * cdb788b redesign reports view
| | * 24bf321 redesign payments view and receipt popup
| | * f5cb721 redesign invoices view
| | * a4fcd1d redesign users and apartments views
| | * d540bb0 add design tokens and redesign login + main shell with sidebar nav and dashboard
| |/
| *   e7ead64 sprint 4: student ID headers, README, requirements, README.txt
| |\
| | * d17d55e update sprint 4 status - readme.txt done, checklist current
| | * 42d7b8c add README.txt with run instructions and test accounts
| | * 8aa0ba7 add student ID headers to all source and test files
| |/
| *   7035fa3 Merge pull request #28 from OmarFrouh/feature/testing-sprint3
| |\
| | * 63f3d4f Fix (Omar review): add 16th test + remove unused PaymentManager import in test_numair_sprint3.py
| | * d98d680 Sprint 3 (Numair): edge-case tests for complaints, update tenant, notice period (16 tests)
| |/
| * 09203d2 Docs (Omar): mark Ahmed Sprint 3 complete
| *   ed1acf0 Merge pull request #27 from OmarFrouh/feature/finance-sprint3
| |\
| | * 82e2573 Sprint 3 (Ahmed): billing receipt popup after payment, get_receipt_data(), 8 tests
| |/
| *   e7efbf1 Merge pull request #26 from OmarFrouh/feature/tenant-sprint3
| |\
| | * 0c884c7 Fix (Omar review): revert payment_manager.py — belongs to Ahmed's branch
| | * e848397 Implement Sprint 3 features: edit tenant, early termination dictionary return, and complaints logic
| * | 0083a87 Docs (Omar): mark Feras Sprint 3 complete
| |/
| * 88d4b0e Sprint 3 (Omar): complaints table, seed data, Sprint 3 test suite (36 tests)
| * 23b35fe update project log - Sprint 2 complete, Sprint 3 roadmap
(venv) → ASD-UWE-2026 git:(main) ✗
```

Appendix D — Contributor Stats

```
   106  Omar Al Frouh <omarfrouh@gmail.com>
     3  Numair <numair@student.uwe.ac.uk>
     3  Omar Alfrouh <omarfrouh@gmail.com>
     2  Ahmed Deraz <asser@Mac.rocket-fibre.home>
     2  Feras <122075868+PhoeniFeras@users.noreply.github.com>
     2  Feras <Feras@ferass-mbp.home>
     1  Ahmed Deraz <asser@MacBook-Air.local>
     1  AiHmeId <ahmed2.deraz@live.uwe.ac.uk>
     1  Feras <Feras@Ferass-MacBook-Pro.local>
(END)
```

Appendix E — Automated Test Run

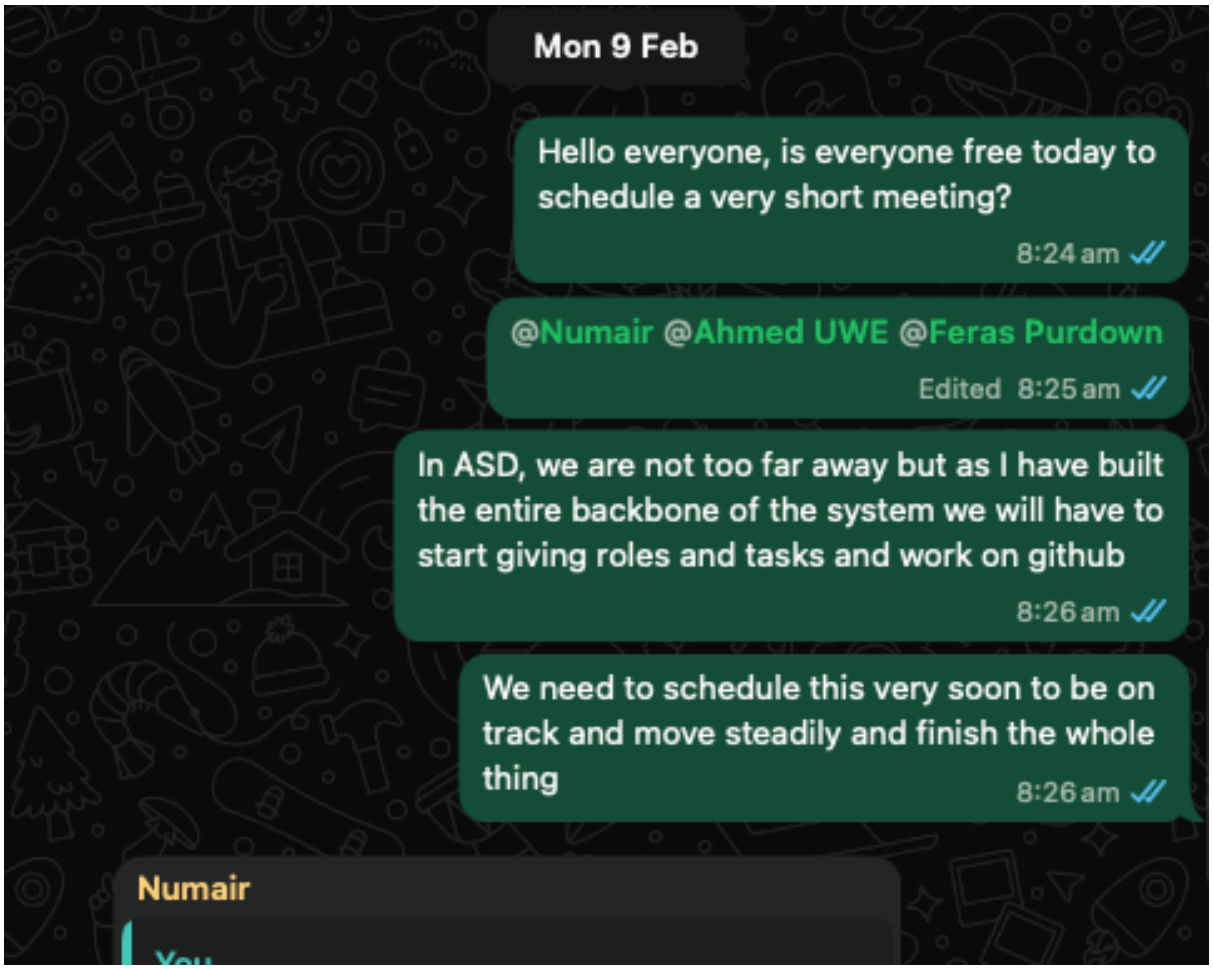
```
test_update_terminated_tenant_fails (test_sprint3.TestUpdateTenant.test_update_terminated_tenant_fails) ... ok
test_archive_tenant_clears_apartment_ghost (test_tenant_manager.TestTenantManager.test_archive_tenant_clears_apartment_ghost) ... ok
test_assign_occupied_apartment_fails (test_tenant_manager.TestTenantManager.test_assign_occupied_apartment_fails) ... ok
test_early_termination_charges_penalty (test_tenant_manager.TestTenantManager.test_early_termination_charges_penalty) ... ok
test_register_and_assign_tenant (test_tenant_manager.TestTenantManager.test_register_and_assign_tenant) ... ok
test_reactivate_already_active_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_already_active_raises) ... ok
test_reactivate_archived_becomes_active (test_tenant_reactivate.TestTenantReactivate.test_reactivate_archived_becomes_active) ... ok
test_reactivate_nonexistent_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_nonexistent_raises) ... ok
test_reactivate_terminated_raises (test_tenant_reactivate.TestTenantReactivate.test_reactivate_terminated_raises) ... ok

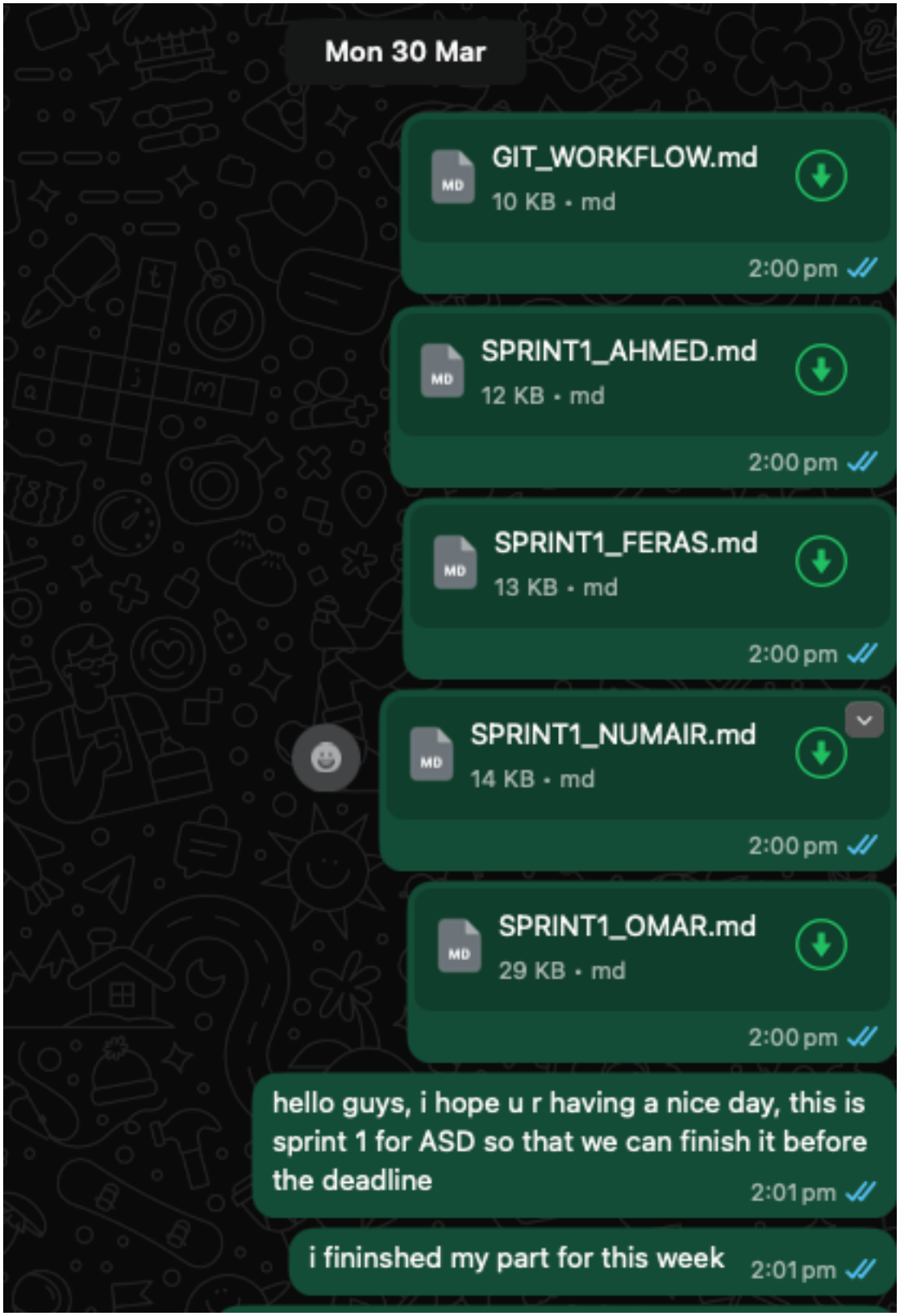
----------------------------------------------------------------------
Ran 330 tests in 75.027s

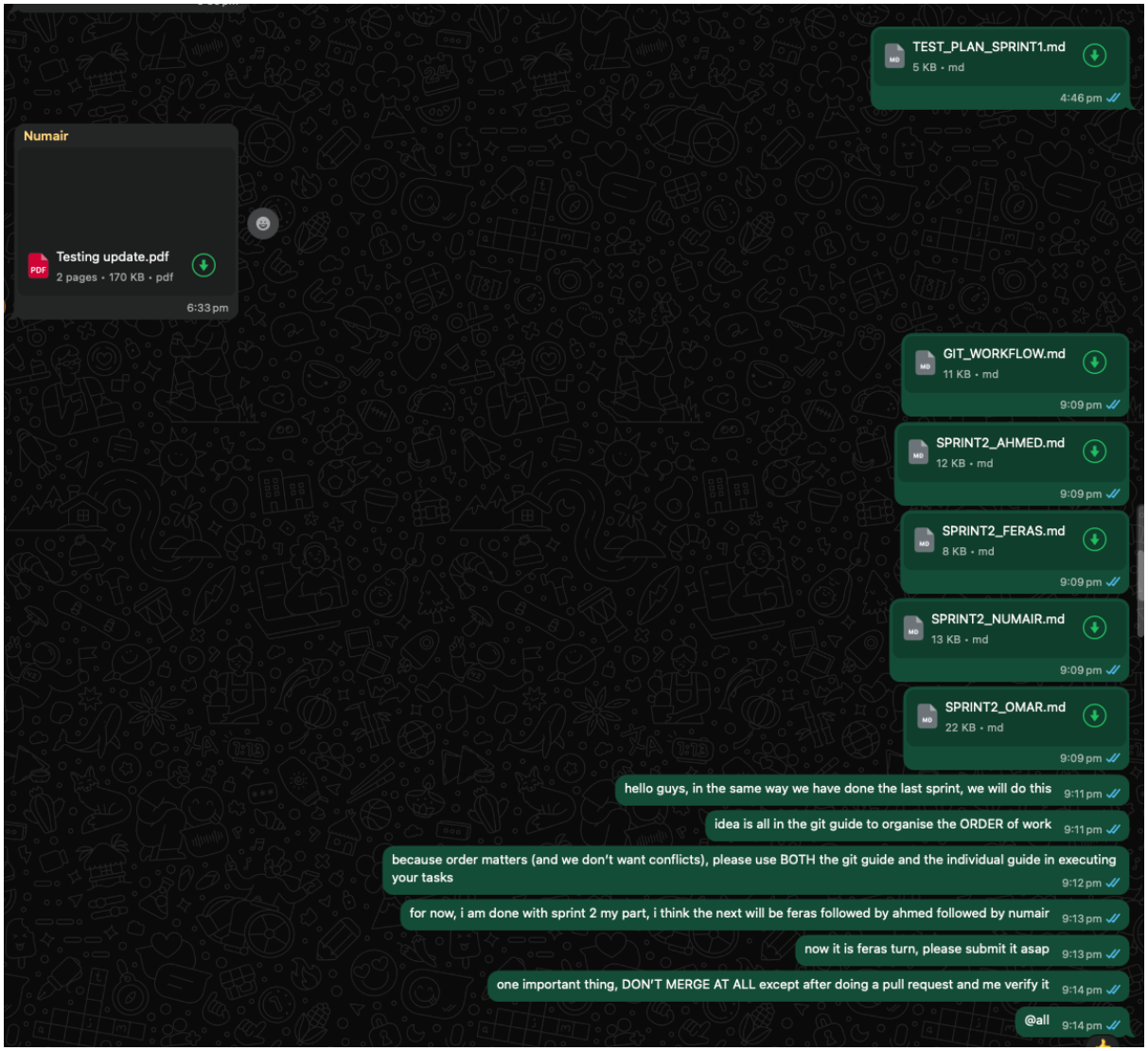
OK
(venv) → ASD-UWE-2026 git:(main) ✗
```

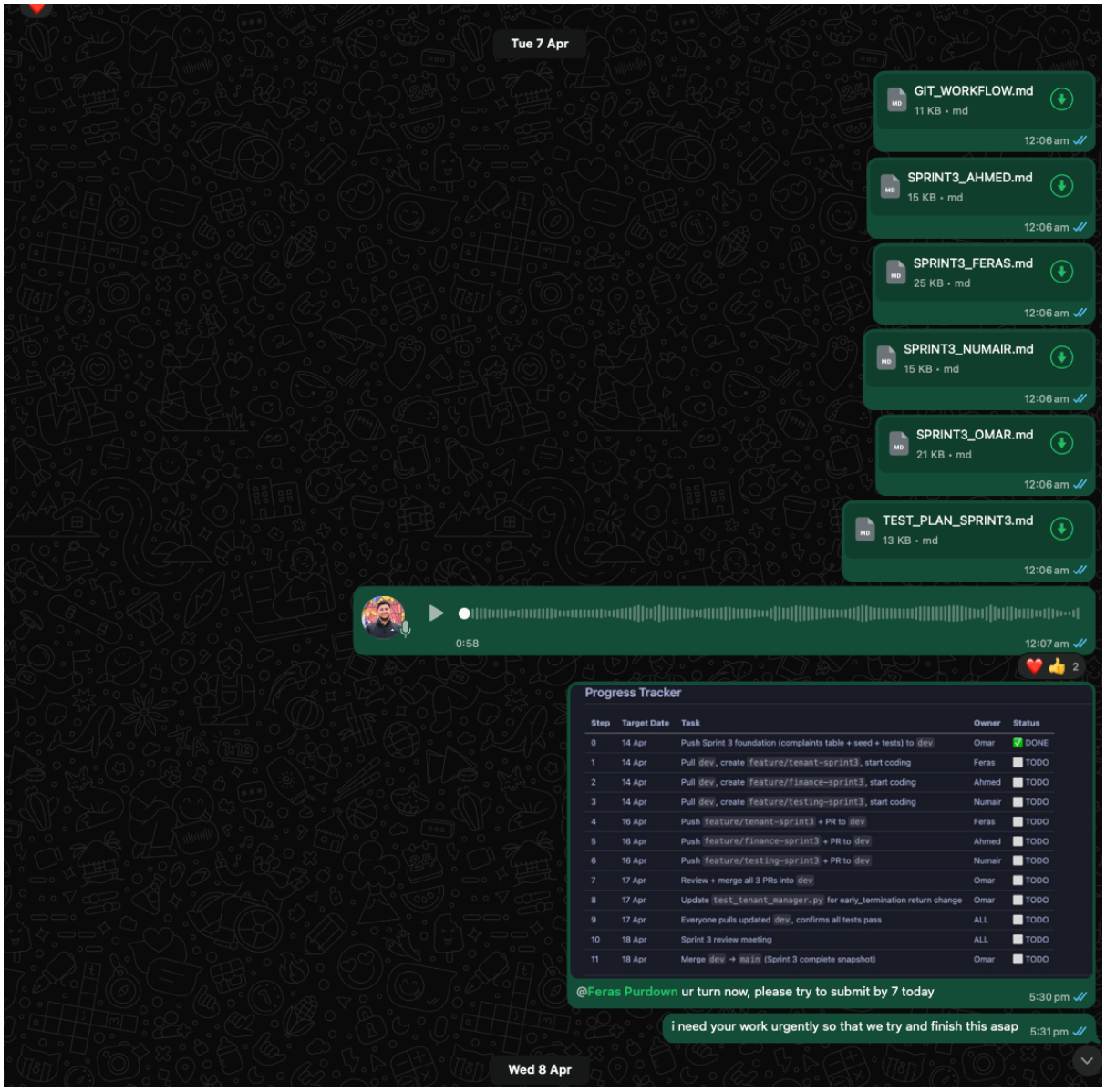
Appendix F — Sprint Planning Artefacts

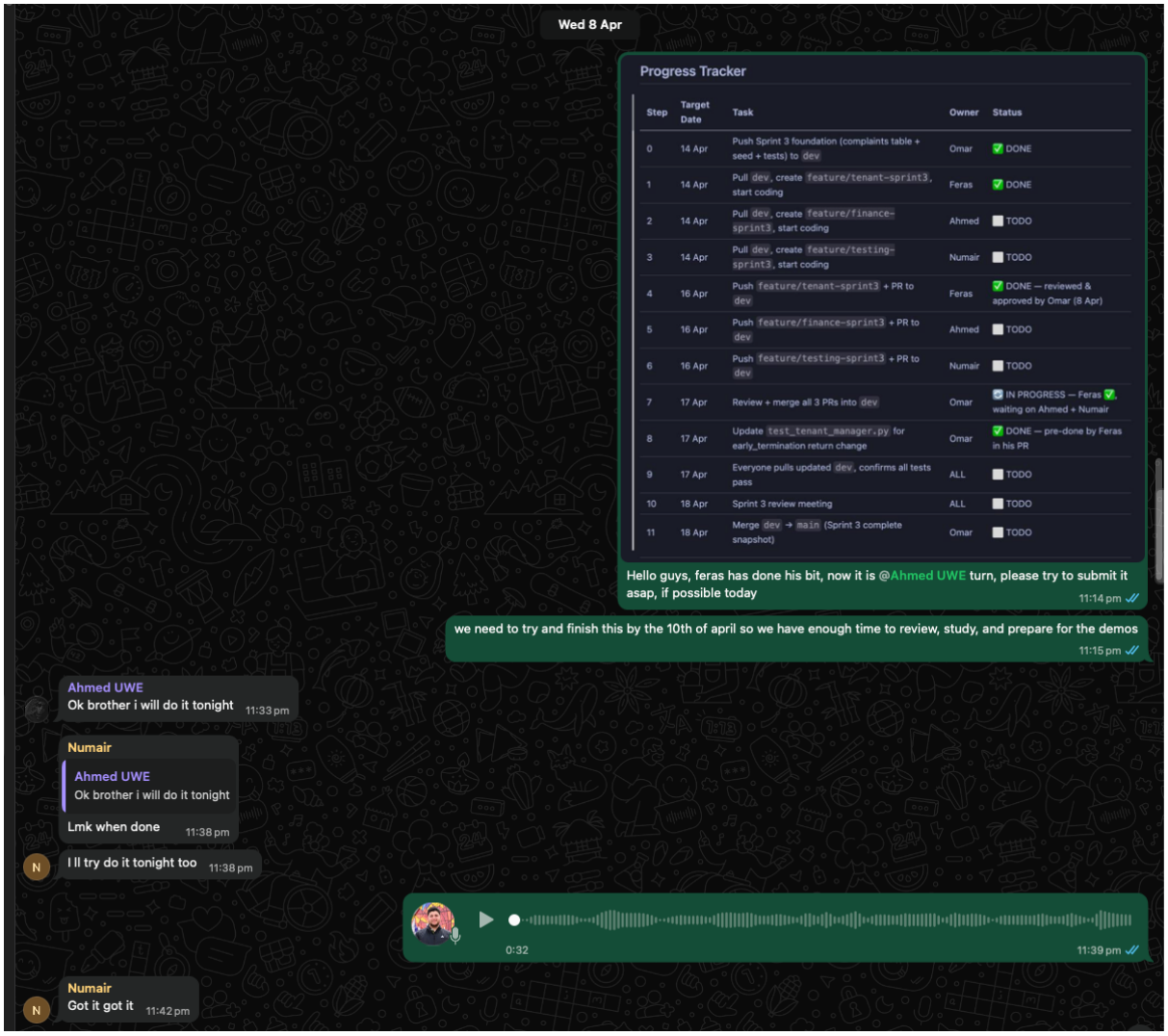
Sprint planning was conducted in person at the start of each sprint. Task assignments, module contracts, and QA checklists were shared with the team via the WhatsApp group and confirmed at the sprint planning meeting. The GitHub Project Board cards and PR history (Appendix C) provide the timestamped record of what was planned and when it was delivered.




Mon 30 Mar
GIT_WORKFLOW.md
10 KB • md
2:00 pm
SPRINT1_AHMED.md
12 KB • md
2:00 pm
SPRINT1_FERAS.md
13 KB • md
2:00 pm
SPRINT1_NUMAIR.md
14 KB • md
2:00 pm
SPRINT1_OMAR.md
29 KB • md
2:00 pm
hello guys, i hope u r having a nice day, this is sprint 1 for ASD so that we can finish it before the deadline
2:01 pm
i fininshed my part for this week
2:01 pm


TEST_PLAN_SPRINT1.md
5 KB • md
4:46 pm
Numair
Testing update.pdf
2 pages • 170 KB • pdf
6:33 pm
GIT_WORKFLOW.md
11 KB • md
9:09 pm
SPRINT2_AHMED.md
12 KB • md
9:09 pm
SPRINT2_FERAS.md
8 KB • md
9:09 pm
SPRINT2_NUMAIR.md
13 KB • md
9:09 pm
SPRINT2_OMAR.md
22 KB • md
9:09 pm
hello guys, in the same way we have done the last sprint, we will do this 9:11 pm
idea is all in the git guide to organise the ORDER of work 9:11 pm
because order matters (and we don't want conflicts), please use BOTH the git guide and the individual guide in executing your tasks 9:12 pm
for now, i am done with sprint 2 my part, i think the next will be feras followed by ahmed followed by numair 9:13 pm
now it is feras turn, please submit it asap 9:13 pm
one important thing, DON'T MERGE AT ALL except after doing a pull request and me verify it 9:14 pm
@all 9:14 pm


Tue 7 Apr
GIT_WORKFLOW.md
11 KB • md
12:06 am
SPRINT3_AHMED.md
15 KB • md
12:06 am
SPRINT3_FERAS.md
25 KB • md
12:06 am
SPRINT3_NUMAIR.md
15 KB • md
12:06 am
SPRINT3_OMAR.md
21 KB • md
12:06 am
TEST_PLAN_SPRINT3.md
13 KB • md
12:06 am
0:58
12:07 am
2
Progress Tracker
Step Target Date Task Owner Status
0 14 Apr Push Sprint 3 foundation (complaints table + seed + tests) to dev Omar DONE
1 14 Apr Pull dev, create feature/tenant-sprint3, start coding Feras TODO
2 14 Apr Pull dev, create feature/finance-sprint3, start coding Ahmed TODO
3 14 Apr Pull dev, create feature/testing-sprint3, start coding Numair TODO
4 16 Apr Push feature/tenant-sprint3 + PR to dev Feras TODO
5 16 Apr Push feature/finance-sprint3 + PR to dev Ahmed TODO
6 16 Apr Push feature/testing-sprint3 + PR to dev Numair TODO
7 17 Apr Review + merge all 3 PRs into dev Omar TODO
8 17 Apr Update test_tenant_manager.py for early_termination return change Omar TODO
9 17 Apr Everyone pulls updated dev, confirms all tests pass ALL TODO
10 18 Apr Sprint 3 review meeting ALL TODO
11 18 Apr Merge dev → main (Sprint 3 complete snapshot) Omar TODO
@Feras Purdown ur turn now, please try to submit by 7 today
5:30 pm
i need your work urgently so that we try and finish this asap
5:31 pm
Wed 8 Apr





End of Report